# Robust RAG under Unreliable Evidence: A Comparability Framework and Evidence Audit

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## Abstract

Robust retrieval-augmented generation (RAG) grounds language models in external evidence, but that evidence can be irrelevant, distracting, conflicting, stale, insufficient, or adversarially corrupted. A growing literature reports progress on "noise robustness", "faithfulness", "security", and "calibration", yet these results are hard to compare: papers that share a label often differ in their evidence assumptions, interventions, metrics, and—crucially—the action the system is expected to take. We argue that robust-RAG progress should be read through a comparability lens, and we make that lens precise with a five-coordinate tuple: failure type, evidence assumption, intervention point, metric target, and action policy. The *action* coordinate—answer, abstain, re-retrieve, filter, verify, expose conflict, clarify, quarantine, trace, or recover—is largely absent from existing RAG, trustworthy-RAG, and RAG-security taxonomies, yet it is what separates a calibrated confidence score from a deployable control policy. Against a 342-row adjudicated, venue-stratified evidence map, we use the tuple to classify comparisons as valid, partial, invalid, or under-specified; we give worked re-analyses where it changes the recommendation, a venue-weighted sensitivity check, an exploratory human validation on hard comparison pairs, and a reporting checklist for future robust-RAG papers. The contribution is a comparability framework, an empirical evidence audit, and a reusable resource for interpreting robust-RAG results under unreliable evidence.

## 1 Introduction: A Comparability Problem in Robust RAG

Retrieval-augmented generation (RAG) combines a parametric language model with external evidence retrieved at inference time (Lewis et al., 2020). Under unreliable evidence, this pipeline can fail because retrieved passages are irrelevant, insufficient, conflicting, stale, unsupported by citations, or adversarially corrupted. The literature has responded with filters, rerankers, note-taking prompts, conflict-aware generators, poisoning defenses, calibration methods, abstention policies, and evaluation toolkits. Yet the resulting claims are often hard to compare.

The core problem is not terminology alone. Two papers can both report "robustness to noisy retrieval" while one changes context construction, another trains the generator, and a third introduces a diagnostic benchmark. Two security papers can both report lower attack success while assuming different attacker access, objectives, and defense actions. A calibration method can improve confidence estimates without deciding whether to answer, retrieve again, or abstain. If these papers are read as substitutable methods, a survey can recommend the wrong benchmark, method, or deployment policy.

This paper asks one question: **when do robust-RAG comparisons fail, and how can we make the missing comparison fields explicit?** We answer with a comparability tuple, apply it to a 342-row adjudicated evidence map, and show five high-value re-analyses where the tuple changes the conclusion. Table 1 previews the practical risk: common survey groupings can induce misleading comparisons when evidence-state and action fields are left implicit.

This prior-survey audit is deliberately conservative. It does not claim that the cited surveys are wrong; it asks whether their broad organizing units are sufficient for direct comparison. In the initial 12 grouping claims we audited (summarized in Appendix Table 7), none provided enough information for a direct leaderboard-style comparison without adding tuple fields, but those counts are an audit sample rather than a field-level estimate.

**Contributions.** We make three contributions. First, a **comparability framework**: a five-

| Audited adjacent-survey grouping | Misleading comparison it can induce | Why the tuple narrows it | Safer recommendation |
|---|---|---|---|
| General/evaluation RAG surveys group retriever, generator, and evaluation work as robustness enhancements (Gao et al., 2024; Sharma, 2025; Yu et al., 2024a). | Pick the best "noise robustness" method across IrrelCtx, RAAT, Chain-of-Note, Sufficient Context, and RGB. | These papers target filtering, training, generator reading, sufficiency estimation, and diagnostic benchmarking under different actions. | Choose by evidence state and action: filter hard negatives, train through noise, read documents, abstain/re-retrieve, or benchmark abilities. |
| Evaluation and trustworthy-RAG surveys group faithfulness metrics, support labels, and conflict methods under factuality or grounding (Yu et al., 2024a; Zhou et al., 2024; Ni et al., 2025). | Use any faithfulness metric as a proxy for conflict robustness. | LLM-judge scores, span labels, and source-conflict reasoning measure different units and reward different actions. | Report answer correctness, claim support, citation entailment, source disagreement, and action under conflict separately. |
| Security-focused surveys group poisoning, prompt injection, data extraction, and defenses as RAG attacks/defenses (Xu et al., 2026; Ni et al., 2025). | Deploy one "RAG defense" expected to reduce all attacks. | Corpus poisoning, memory poisoning, data extraction, bounded corruption, and instruction hijacking assume different attacker capabilities and objectives. | Match defenses to a capability envelope and state whether they prevent, demote, detect, trace, recover, or limit leakage. |
| General and evaluation surveys group adaptive retrieval, uncertainty, calibration, and abstention under adaptive or reliable RAG (Sharma, 2025; Yu et al., 2024a). | Treat confidence calibration as deciding when to retrieve or abstain. | Confidence target, evidence sufficiency, retrieval trigger, action set, and cost are separable. | Evaluate policies over answer, retrieve, verify, clarify, abstain, and escalate with coverage and cost. |
| Broad RAG surveys often treat multimodal, graph, table, domain, and agentic RAG as application extensions (Gao et al., 2024; Sharma, 2025). | Transfer passage-level citation or recall metrics to pages, tables, graphs, tools, and images. | The evidence object changes; layout, cell provenance, graph edge, visual region, and domain authority become validity conditions. | Use beyond-text papers as typed-evidence stress tests unless running a separate systematic multimodal/agentic survey. |

Table 1: Concrete delta relative to audited adjacent-survey groupings. The table uses survey-family organizing units as the starting point, then shows how recommendations narrow when robust-RAG papers are recoded by failure, assumption, intervention, metric, and action.

coordinate tuple that separates the evidence state, intervention point, and metric target from—distinctively—the *action policy* a system should follow, and that labels any two-paper comparison as valid, partial, invalid, or under-specified. The action coordinate is largely absent from existing RAG, trustworthy-RAG, and security taxonomies, yet it is what makes a robustness number portable. Second, an **empirical evidence audit**: a reusable adjudicated, venue-stratified evidence map—Map A is a 268-row double-coded seed, and Map B extends it to 342 rows after independent second coding and adjudication of 148 candidate additions—released with its reliability, override, and provenance trail. Third, an **applied meta-analysis**: worked re-analyses where the tuple changes the recommendation, a venue-weighted sensitivity check, an exploratory human validation, and a concrete reporting checklist that translate the framework into benchmark design, method choice, and deployment policy.

## 2 A Comparability Framework: The Five-Coordinate Tuple

We read each robust-RAG paper as a tuple:

$$(\text{failure}, \text{assumption}, \text{intervention}, \text{metric}, \text{action}).$$

**Failure** is the evidence problem: noisy retrieval, distraction, knowledge conflict, adversarial corruption, OOD/uncertainty, calibration/control, or evaluation. **Assumption** records the evidence state or threat model: benign hard negatives, insufficient context, stale or conflicting sources, attacker-created documents, instruction-bearing retrieved text, or typed evidence beyond passages. **Intervention** locates the method in the pipeline: corpus/KB, retriever/context, context assembly, generator, system control, or evaluation. **Metric** records what is optimized: answer correctness, support, faithfulness, attack success, selective risk, calibration, action utility, or cost. **Action** records what the system should do: answer, abstain, retrieve again, filter, verify, expose conflict, clarify, quarantine, trace, or recover.

A comparison is *valid* only when these coordinates align closely enough for the reported metric to answer the same question. It is *partial* when the papers are complementary mechanisms inside the same problem family. It is *invalid* when a shared label hides different scientific games. It is *under-specified* when the paper omits a coordinate needed to decide. Appendix Table 8 gives the full codebook; the main point is simple: a score is not portable unless the evidence state, metric target,

and action policy are portable.

**Relation to existing surveys.** Existing RAG surveys organize the field by pipeline stage (Gao et al., 2024; Sharma, 2025), by trust dimension (Zhou et al., 2024; Ni et al., 2025), by security surface (Xu et al., 2026), or by evaluation artifact (Yu et al., 2024a). These are valuable maps of *what exists*, but they are organized for coverage rather than comparability: two methods can share a pipeline stage, trust axis, threat label, or benchmark and still answer different scientific questions. Our tuple is orthogonal—it does not replace these axes but supplies the comparison fields they leave implicit. In particular, the *action* coordinate (answer, abstain, re-retrieve, filter, verify, expose conflict, clarify, quarantine, trace, recover) is under-represented in prior taxonomies, yet it is what separates a calibrated confidence score from a deployable control policy, and a diagnostic benchmark from a mitigation method. Table 2 states this delta coordinate by coordinate: across the five re-analysis families below, ordinary labels permit a direct comparison while tuple coding isolates a breaking coordinate every time, most often the action.

We now read the literature through this lens, mapping the evidence base onto the tuple coordinates by primary failure mode and intervention point. This is an organizing pass over an adjudicated evidence map, not an exhaustive survey: many papers span more than one branch, and we flag where the comparability tuple separates superficially similar work. The 342-row adjudicated evidence map (Map B) behind this analysis is summarized in Section 10 and detailed in Appendix E.

## 3 Evidence-Centric Robustness: Noisy Retrieval and Distraction

The first branch of robust RAG studies what enters the context. Early work on irrelevant context established that retrieval errors do not merely lower recall; they can actively redirect generation (Yoran et al., 2024). Noise-aware training and robust RALM work then separated random negatives, hard negatives, counterfactual passages, and noisy retrieved evidence as different stressors rather than one generic perturbation (Fang et al., 2024; Park and Lee, 2024; Ding et al., 2025). Benchmark papers such as RGB and URAG made this decomposition operational by testing negative rejection, information integration, counterfactual evidence, and uncertainty under retrieval noise (Chen et al., 2024a; Nguyen et al., 2026). These works show that "noise robustness" is not a single axis but a family of distinct stressors.

Distraction work focuses on negative marginal utility: a passage is harmful not simply because it is irrelevant, but because it changes the answer, confidence, attribution, or reasoning path. Long-context and context-utility studies show that more context can increase the probability of including plausible but misleading passages (Gao et al., 2026b; Jin et al., 2024a; Hagström et al., 2025). Utility-oriented retrieval and downstream-aware metrics therefore ask whether a passage helps the generator rather than whether it only matches the query (Trappolini et al., 2025; Zhu et al., 2024; Dai et al., 2026; Niu et al., 2026; Gao et al., 2025b). Context compression and filtering methods try to remove irrelevant spans or documents while preserving evidence needed for multi-hop reasoning (Wang et al., 2023b; Yu et al., 2023; Joren et al., 2025; Hong et al., 2023; Cho et al., 2024; Zeng et al., 2025b).

The later evidence-centric literature broadens noise beyond the top-$k$ ranking to query phrasing (Percin et al., 2025), context ordering and length (Kim et al., 2026; Gabín et al., 2026; Pandey, 2026; Wang et al., 2025d; Chen et al., 2025b; Liu et al., 2025d), and pipeline-level construction, compression, and retrieval defects (Chen et al., 2025c; Yang et al., 2025; Kachuee et al., 2025; Chakraborty et al., 2025; Oh et al., 2025; Tan et al., 2025; Kim, 2025), with recent systems and benchmarks adding stress tests for retrieval defects, context utility, and robust fine-tuning (Jin et al., 2025; Iratni et al., 2025; Sui et al., 2025; Qiao et al., 2024; Wang et al., 2024; Zhang et al., 2024a; Qiu et al., 2024; Park et al., 2024; Abdumalikov et al., 2024; Kim et al., 2024; Xu et al., 2023). The common lesson is that retrieval should be evaluated as context construction, not recall.

Venue-accepted work reinforces the trend—efficient RAG and reranking (Hofstätter et al., 2023; Chen et al., 2025f; Wang et al., 2025g), graph- and knowledge-guided retrieval (He et al., 2024; Luo et al., 2025; Ma et al., 2025; Yu et al., 2025a; Hu et al., 2025; Li et al., 2025d), long-context benchmarks (Xu et al., 2025; Li et al., 2025c), and compression and retrieval-utility methods (Cao et al., 2024; Cheng et al., 2024b; Chirkova et al., 2025; Yu et al., 2024b; Zeng et al., 2025a; Dai et al., 2025a; Hsia et al., 2025; Chen et al., 2025a; Wei et al., 2025)—all treating retrieval as evidence construction rather than recall.

| Comparison family | Ordinary taxonomy code | Five-coordinate code adds | Error reduced |
|---|---|---|---|
| Noise methods | Same failure family: noisy retrieval / distraction. | Intervention separates filter, training, reader, sufficiency predictor, and benchmark; action separates answer, filter, abstain, and re-retrieve. | Prevents ranking complementary pipeline components as substitutes. |
| Faithfulness/conflict | Same trust/factuality or grounding family. | Metric unit separates LLM-judge score, span support, answer correctness, and source-conflict resolution; action records expose/resolve/answer. | Prevents treating citation support as conflict robustness. |
| Conflict vs. poisoning | Same broad "conflicting evidence" or robustness family. | Evidence assumption separates natural disagreement from attacker-created evidence; metric separates reconciliation from attack success. | Prevents transfer from benign conflict methods to adversarial defenses. |
| Security defenses | Same security/attack-defense family. | Assumption records attacker access, insertion budget, objective, and defense action. | Prevents pooling prevention, demotion, detection, traceback, recovery, and leakage control. |
| Calibration/control | Same uncertainty/adaptive RAG family. | Action policy separates confidence estimation, retrieval trigger, abstention, clarification, verification, and escalation. | Prevents treating better calibration as a complete action policy. |

Table 2: Side-by-side delta between ordinary robust-RAG taxonomy coding and the five-coordinate tuple. In the five main re-analyses, ordinary family labels would allow a direct or ambiguous comparison in all five cases; tuple coding identifies a breaking coordinate in all five and changes the recommendation.

## 4 Model-Centric Robustness: Conflict, Faithfulness, and Attribution

Once evidence enters the prompt, robustness depends on how the generator uses it. Knowledge-conflict papers study contradictions between parametric memory and retrieved evidence, or among retrieved sources (Jin et al., 2024b; Zhang et al., 2025f; Wang et al., 2025a; Xie et al., 2023). These papers show that a model can over-trust retrieved evidence, resist correct retrieval because of parametric priors, or silently choose one source without exposing disagreement. Conflict-aware methods therefore localize discrepancies, compare internal and external knowledge, or introduce source-aware reasoning before generation (Ma et al., 2026; Zhou et al., 2025a; Si et al., 2025; Huang et al., 2025a; Lee et al., 2025; Cattan et al., 2025; Hou et al., 2024; Huang et al., 2024).

Faithfulness and attribution papers study whether generated claims are supported by evidence. RAGTruth and related grounding work provide fine-grained hallucination or support labels, while source-preference and entity-conflict studies examine whether models cite and trust the right evidence (Niu et al., 2024; Pham et al., 2024; Kortukov et al., 2024; Zhang et al., 2024b; Schuster et al., 2026). These papers connect robust RAG to citation verification: document-level citation is often too coarse, because a document can contain multiple claims, qualifiers, and contradictions. Claim-level or sentence-level attribution is more informative but increases annotation and verification cost (Zhao et al., 2026a; Shin et al., 2026; Lin et al., 2025b; Choi et al., 2025b; Wu et al., 2024; Dai et al., 2025b; Ge et al., 2025).

The broader map extends this to source reliability, contextual faithfulness, real-document conflicts, and hidden-state or logit interventions (Jin et al., 2024c; Wang et al., 2023a; Li et al., 2026b; Shihab et al., 2026; Ju et al., 2026; Liu et al., 2025c; Yan et al., 2025; Huo et al., 2025), and to richer benchmarks, source-aware generation, and practical pipelines (Wu et al., 2025b; Long et al., 2025; Jia et al., 2025b; Wang et al., 2025b; Xu et al., 2024; Li et al., 2024a; Yang et al., 2024; Marjanović et al., 2024; Tan et al., 2024a; Zhang et al., 2023). One distinction recurs: evidence can be present but ignored, cited but unsupported, or correct but less trusted than a stale parametric belief.

Venue-accepted faithfulness work adds the same distinction: grounded attribution and refusal (Song et al., 2025b), context-faithfulness benchmarks under misleading evidence (Ming et al., 2025), hallucination and mechanistic detectors (Sun et al., 2025b; Sriramanan et al., 2024), and parametric–contextual knowledge reconciliation (Zhao et al., 2025) together show that robust RAG needs both evidence-use diagnostics and source-reliability judgments. Older grounded-generation work (Qian et al., 2023; Hofstätter et al., 2023) studies the same evidence–text interface, motivating treatment of early retrieval-augmented generation as a foundation rather than a 2025–2026 preprint phenomenon.

## 5 Adversarial Evidence Corruption and Security

Adversarial RAG work asks what happens when evidence unreliability is created by an adaptive

attacker. Corpus and retrieval poisoning papers show that small numbers of malicious passages can manipulate dense retrieval or steer final answers (Zhong et al., 2023; Zou et al., 2025; Xue et al., 2024; Geng et al., 2025; Zhang et al., 2025a). Certified and aggregation-based defenses study robustness under bounded corruption assumptions (Xiang et al., 2024; Cheng et al., 2025; Kim et al., 2025a), while reranking and detection defenses use gradient, attention, clustering, or anomaly signals to demote suspicious evidence (Yin et al., 2026; Jin et al., 2026; Dekel et al., 2026; Choudhary et al., 2025; Si et al., 2025).

Prompt-injection and backdoor work targets a different interface. Instead of inserting false facts, the attacker may insert instructions that the model treats as commands, or triggers that activate malicious behavior through retrieved contexts (Chaturvedi et al., 2025; Chang et al., 2026; Cheng et al., 2024a; Chaudhari et al., 2024; Song et al., 2026; Kim et al., 2025b). These attacks require retrieval realism: a malicious document must be retrieved under plausible queries, not only under attacker-crafted prompts. This is why we separate factual poisoning, indirect prompt injection, backdoor triggers, and prompt-template attacks as different threat games even when they all appear as retrieved text.

The security map is broad and fast-moving, spanning poisoning benchmarks and arenas (Liang et al., 2025b; Chen et al., 2025e; Zhang et al., 2025d; Tan et al., 2024b), traceback and forensics (Zhang et al., 2025c,b), privacy and membership inference (Li et al., 2026a; Gao et al., 2026a; Song et al., 2025a; Shereen et al., 2025), opinion or fairness manipulation (Chen et al., 2025i; Bagwe et al., 2025), graph, multimodal, or reasoning-style poisoning surfaces (Liang et al., 2025a; Wen et al., 2025; Zhou and Wang, 2025), and many preprint defenses and attacks (Pathmanathan et al., 2025; Jiao et al., 2025; Li et al., 2025b; Chen et al., 2025h; Xian et al., 2025b; Choi et al., 2025a; Stefano et al., 2024; Suo et al., 2025; Edemacu and Shokri, 2026; Jia et al., 2025a; Lin et al., 2025a; Chen et al., 2024d). Its use is not to rank attacks but to motivate capability-normalized reporting.

Venue-accepted security work extends the surface to agent and memory poisoning, data extraction, medical-domain risk, robust web search, and semantic conflict-free filtering (Chen et al., 2024e; Xian et al., 2025a; Shen et al., 2025; Si et al., 2025; Zhang et al., 2025e); these are comparable only once the attacker's control over corpus, prompt, query, memory, or retrieved evidence is specified, which is exactly what threat-model normalization enforces.

## 6 System Control: Calibration, Abstention, and Adaptive Retrieval

System-control work converts evidence judgments into actions. Answerability and sufficient-context papers ask whether the retrieved context contains enough evidence to answer (Joren et al., 2025; Peng et al., 2024; Rosenthal et al., 2026; Liu et al., 2025a). Uncertainty and calibration work asks whether confidence reflects answer correctness, evidence sufficiency, retrieval usefulness, or selective risk (Jang et al., 2025a; Li et al., 2024b; Nguyen et al., 2026; Liu et al., 2026b; Soudani et al., 2025a; Qiu et al., 2026). Adaptive retrieval work asks when the system should retrieve, retrieve again, or answer from internal knowledge (Asai et al., 2024a; Labruna et al., 2024; Yan et al., 2024; Liu et al., 2025b; Moskvoretskii et al., 2025).

The control literature highlights a recurring failure: confidence is not the same as evidence sufficiency. A model can be confident in an unsupported answer, or uncertain even when evidence is present but hard to integrate. Trust-or-abstain and self-aware RAG methods therefore model whether to answer, abstain, retrieve more, or ask for clarification (Zhu et al., 2026; Xie et al., 2026; Chen et al., 2024b; Soudani et al., 2025b; Sun et al., 2025a; Chen et al., 2026b; Zhao et al., 2026b; Deiseroth et al., 2025; Jia et al., 2026; Fadeeva et al., 2025; Huang et al., 2025b). Multi-turn and underspecified-query benchmarks add another layer: the correct action may be clarification before retrieval, not direct answering (Rosenthal et al., 2026; Bazarova et al., 2026; Chen et al., 2026a).

Recent work extends control to cost-aware retrieval, conformal or set-valued prediction, risk-sensitive abstention, and adaptive policies over multiple retrieval steps (Zhou et al., 2026; Sinha, 2025; Wu et al., 2025a; Zubkova et al., 2025; Wang et al., 2025f; Chen et al., 2025d; Li et al., 2025a; Marina et al., 2025; Foo et al., 2025; Shankar et al., 2025; Huang et al., 2025c; Min et al., 2025). These papers motivate an action-control view: robust RAG is not only about reducing errors, but about choosing the right response under uncertainty and cost.

Venue-accepted control work makes the action view explicit: calibration framed as retrieval-and-

answer control (Jang et al., 2025b), learned adaptive retrieval and critique (Asai et al., 2024b), query routing across systems or long-context processing (Jin et al., 2024a; Xu et al., 2025; Li et al., 2025c), multi-step chain-of-retrieval (Wang et al., 2025c; Tang et al., 2024), and feedback- or RL-driven policies for when to retrieve, rewrite, and answer (Gao et al., 2025a; Feng et al., 2025; Zhou and Chen, 2025; Li et al., 2025e). Action control is thus becoming a central design axis in venue-published work, not only a preprint agenda.

## 7 Evaluation and Benchmarks

Evaluation papers provide the diagnostic infrastructure for robust RAG. RGB decomposes RAG reliability into multiple abilities (Chen et al., 2024a); RAGTruth provides fine-grained hallucination annotations (Niu et al., 2024); query-robustness and context-utility work stress retrieval interfaces (Percin et al., 2025; Trappolini et al., 2025); security benchmarks and arenas evaluate poisoning and defense assumptions (Liang et al., 2025b; Chen et al., 2025e; Tan et al., 2024b); and uncertainty benchmarks evaluate whether systems know when their answers are reliable (Nguyen et al., 2026; Qiu et al., 2026). Survey papers on RAG, trustworthy RAG, secure RAG, and evaluation provide the broader landscape (Gao et al., 2024; Sharma, 2025; Yu et al., 2024a; Zhou et al., 2024; Ni et al., 2025; Xu et al., 2026; Mu et al., 2026; Liu et al., 2026a).

Accepted benchmark work broadens this infrastructure with grounding and hallucination verification (Leemann et al., 2025; Sun et al., 2025b; Sriramanan et al., 2024) and with multimodal or graph RAG benchmarks (Wang et al., 2025e; He et al., 2024; Luo et al., 2025; Li et al., 2025f), which we treat as boundary and future-direction evidence rather than the empirical backbone for text RAG.

What is still missing is joint evaluation. Existing benchmarks often isolate one failure mode: noise, conflict, poisoning, or unanswerability. Real systems face combinations. A benchmark that includes a stale source, a hard negative, insufficient evidence, and a malicious document will test a different capability from any single-axis benchmark. This motivates the action-trace blueprint: robust-RAG evaluation should score evidence state, answer correctness, action choice, cost, and threat assumptions together.

## 8 Re-analyses That Change Recommendations

Table 3 is our central analysis artifact. Each row names concrete robust-RAG papers that adjacent surveys could reasonably group together, then shows which tuple coordinate blocks the comparison and what recommendation changes. The rows are representative cases from predeclared comparison families rather than a claim that these five exhaust the field. The same families are expanded in Appendix Tables 24 and 26; Appendix Table 28 reports a systematic 60-pair audit over the broader pool, and a blind expert validation tests 60 unseen hard pairs.

**A worked case.** Five papers in the first row of Table 3 are routinely grouped as "noise-robustness" methods, yet the tuple places them on different coordinates: Yoran et al. (2024) and Fang et al. (2024) intervene on *evidence reading* via noise-aware generation, Yu et al. (2023) on *context assembly* via note-taking, Joren et al. (2025) on *system control* via sufficiency-gated abstention, while Chen et al. (2024a) is a *diagnostic benchmark*. A single leaderboard over these conflates a training objective, a prompting strategy, an abstention policy, and a measurement instrument, so a higher number can reflect an easier retriever, not a better method. The tuple's output is thus not "method $X$ wins" but a routing rule keyed on evidence state: filter rankable hard negatives, train through unrankable noise, gate on sufficiency when abstention is allowed, and use the benchmark only to measure. The remaining rows behave the same way: the breaking coordinate, not the shared label, decides whether a comparison is even well posed.

The re-analyses yield a narrower but more actionable synthesis than a citation list: noise robustness decomposes into context construction, generator training, evidence reading, sufficiency prediction, and diagnostic benchmarking; faithfulness and conflict separate span support from source reconciliation; natural conflict and poisoning share annotation needs but not threat games; security defenses are comparable only inside capability envelopes; and calibration, retrieval control, abstention, and clarification are action policies, not interchangeable uncertainty scores.

| Ordinary survey comparison | Five-coordinate diagnosis | Breaking coordinate(s) | Verdict | Changed recommendation |
|---|---|---|---|---|
| Yoran et al. (2024), Fang et al. (2024), Yu et al. (2023), Joren et al. (2025), and Chen et al. (2024a) as "noise robustness." | Shared broad failure label, but filtering/back-off, noise-aware training, generator evidence reading, sufficiency estimation, and ability benchmarking intervene at different points and reward different metrics/actions. | Intervention, metric, action. | Partial / under-specified | Do not pick one "best noise method." Choose by evidence state: hard-negative filtering, training-through-noise, document reading, abstention/re-retrieval, or diagnostic benchmarking. |
| Niu et al. (2024), RAGAS/ARES (Es et al., 2024; Saad-Falcon et al., 2024), Zhang et al. (2025f), and Wang et al. (2025a) as a common faithfulness toolbox. | LLM-judge scores, span-level support labels, and fact/source conflict reasoning measure different units. A supported citation, correct answer, and resolved conflict are not equivalent. | Metric unit, failure subtype, action. | Partial | Report answer correctness, claim support, citation entailment, source disagreement, and action under conflict separately. |
| Zhang et al. (2025f) / Wang et al. (2025a) vs. Zou et al. (2025) as "conflicting evidence." | Natural or diagnostic source conflict differs from attacker-created evidence corruption. Conflict methods ask how to reconcile or expose disagreement; poisoning asks whether an attacker can steer an answer. | Evidence assumption, threat model, metric, action. | Invalid for direct ranking | Share claim/source annotations where useful, but evaluate conflict resolution and adversarial steering under separate threat games. |
| Zou et al. (2025), AgentPoison (Chen et al., 2024e), Spill-Beans (Unknown, 2025), ReliabilityRAG (Shen et al., 2025), and SeCon-RAG (Si et al., 2025) as generic RAG security. | Corpus poisoning, memory poisoning, data extraction, bounded filtering, and semantic conflict-free filtering assume different attacker capabilities and defense actions. | Assumption, objective, defense action. | Partial / invalid across threat models | Report capability envelope, clean utility, and whether the method prevents, demotes, detects, traces, recovers, or limits leakage. |
| Asai et al. (2024a), FLARE/DRAGIN (Jiang et al., 2023; Su et al., 2024), Joren et al. (2025), and Jang et al. (2025a) as adaptive or uncertainty RAG. | Confidence, evidence sufficiency, retrieval trigger, critique policy, and abstention answer different control questions. Calibration can improve belief estimates without selecting the right action. | Failure subtype, metric target, action set. | Partial | Evaluate over a shared action set: answer, retrieve again, verify, clarify, abstain, or escalate, with coverage and cost. |

Table 3: Five worked re-analyses. Each row shows how an ordinary survey grouping can imply a misleading comparison, which tuple coordinate breaks the comparison, and what recommendation changes after normalization.

## 9 Reporting Checklist for Robust RAG

The most important practical consequence is that robust-RAG papers should report enough information for comparison. Table 4 gives the reporting checklist implied by this analysis. It is intentionally concrete: without failure definition, evidence assumption, intervention boundary, retrieval setting, metric target, and action policy, a result cannot be placed in a meaningful comparison set. The action-labeled pilot in the appendix illustrates why action fields are necessary, but it remains a schema sanity check: a correct answer can be the wrong action when evidence is poisoned, stale, or insufficient, and a refusal can be wrong when re-retrieval would recover support. Real RAG/LLM baselines over such action-labeled items are left as future benchmark work.

## 10 Evidence Audit, Sensitivity, and Validation

**Evidence base and reliability.** The audit is grounded in a reusable scoping resource: 1,904 retrieved records, 1,399 deduplicated candidates, and 536 relevant included papers. Map B is a curated, adjudicated subset of 342 of these papers: Map A is a 268-row double-coded seed (from 274 double-coded candidates, six later excluded), extended after independent second coding and adjudication of 148 candidate additions (74 retained, 74 excluded). Its source strata are 126 venue/proceedings-linked rows, 207 arXiv/preprint rows, and 9 workshop/other rows. Second coding of the additions reached $\kappa = 0.72$–$0.77$ across the primary fields and $\kappa \geq 0.74$ against the final adjudicated labels, and 125 of 148 additions changed on at least one field during adjudication—an override rate we report rather than hide (Appendix Tables 41 and 42). We read the map as a comparison-family substrate, not a prevalence estimate—the single caveat governing every map-level number here.

**Venue-weighted sensitivity.** Because 207 of 342 rows are preprints, we ask which conclusions survive when the map is restricted to its 126 venue-linked rows (Appendix Table 48). Three conclusions carry the strongest venue-linked support: retrieval acts as context construction rather than neutral recall (28 venue-linked noisy-retrieval and

| Required field | What to report | If omitted | Example consequence |
|---|---|---|---|
| Failure definition | Noise, distraction, conflict, adversarial corruption, uncertainty/OOD, calibration/control, or evaluation target. | Papers with the same robustness label may address different failures. | Noise training is ranked against sufficiency prediction. |
| Evidence assumption | Gold support availability, distractor hardness/rank, stale/conflicting sources, typed evidence, or attacker capability. | Stressors are not portable. | Natural conflict is compared with poisoning. |
| Intervention boundary | Corpus/KB, retriever/context, context assembly, generator, system control, or evaluation artifact. | Pipeline modules are treated as substitutable. | A benchmark is ranked against a mitigation method. |
| Retrieval setting | Corpus snapshot, retriever, top-$k$, chunking, reranking/compression, and index drift controls. | Results cannot be reproduced or normalized. | A filter looks better because its retriever retrieved easier negatives. |
| Metric target | Answer correctness, support, faithfulness, attack success, selective risk, calibration, cost, or action utility. | Different objectives collapse into one score. | Lower attack success hides clean-utility loss. |
| Action policy | answer, abstain, retrieve again, filter, verify, expose conflict, clarify, quarantine, trace, recover. | Accuracy and confidence are over-interpreted. | A calibrated answerer still answers from insufficient evidence. |
| Uncertainty / abstention handling | Coverage, selective risk, confidence target, abstention reward, and re-retrieval cost. | Abstention can hide retrieval failure. | Refusal is rewarded when re-retrieval should be the correct action. |
| Security threat model | Attacker access, query control, retriever visibility, insertion budget, objective, and adaptive-defense assumptions. | Security numbers are not comparable. | Prompt injection defenses are claimed to protect against corpus poisoning. |

Table 4: Reporting checklist for robust RAG under unreliable evidence. The checklist turns the comparability framework's diagnosis into a concrete reporting norm for future papers and benchmarks.

distraction rows), answer accuracy alone is insufficient for robust-RAG evaluation (28 evaluation/benchmark rows), and RAG security requires capability normalization before attack-success numbers are comparable (26 adversarial-corruption rows). Calibration/action-control and knowledge-conflict conclusions remain conceptually supported but rest on fewer directly comparable venue-linked evaluations, and extension-layer claims are retained only as typed-evidence stress tests.

**Human validation (exploratory).** Two small human studies stress-test the tuple; both are exploratory and directional, not confirmatory. In a broad pilot, 30 independent annotators each judged 12 hard pairs (15 using ordinary judgment, 15 using the tuple), and agreement with a blind external gold panel rose from 0.80 to 0.98. In a harder, held-out re-annotation, three control and three protocol annotators judged 60 unseen pairs against an independent nine-vote gold panel (Fleiss $\kappa = 0.772$): agreement rose from 0.567 to 0.883 ($+0.317$; clustered-bootstrap 95% CI $[0.19, 0.44]$). We take the 60-pair study as primary—unseen, deliberately hard (partial/invalid verdicts), and scored against a stable panel—and the pilot as supporting breadth. Two cautions apply: each 60-pair arm has only three annotators, so we report the direction of the gain, not an effect size; and because the tuple and the gold rubric share coordinates, part of the gain is definitional, reduced but not removed by held-out pairs and a provenance-blinded gold packet. We therefore claim improved triage on hard comparisons, not a calibrated magnitude (Appendix Tables 59 and 60).

**Implications.** For benchmark builders, the implication is to annotate evidence state and action before ranking systems. For method designers, it is to state the operating region: negative type, source reliability, attacker capability, metric target, and action set. For system builders, it is to treat robust RAG as risk-adaptive control rather than a fixed retrieve-then-answer pipeline: an assistant facing a stale or conflicting source should not answer with higher confidence, but re-retrieve, surface the conflict, or abstain, and report that decision and its cost.

## 11 Conclusion

Robust-RAG progress is hard to interpret when papers share labels but not evidence states, interventions, metrics, or actions. A five-coordinate comparability tuple makes these mismatches explicit; applied to an adjudicated evidence map, it changes how common groups of papers should be read: many are complementary, invalid for direct ranking, or under-specified rather than competing leaderboard entries. The next step is not another universal robustness score, but action-labeled evaluation with corpus snapshots, evidence-state labels, gold actions, baseline systems, and error analysis.

## Limitations

Four limitations bound the claims. First, Map B is open-PDF based and scoping, not venue-complete: it improves venue-linked coverage over Map A

(126 versus 42 venue/proceedings-linked rows), but preprints remain the largest stratum and map counts are not field-level rates. Second, the map is deepest for text/document RAG; beyond-text work is used only as typed-evidence stress tests. Third, the human studies are exploratory and validate triage on hard partial/invalid pairs, not the whole map or every verdict category, and the action-labeled pilot is a schema check rather than a real-system benchmark. Fourth, citation auditing is tiered: the papers carrying the framework, the re-analyses, and the venue-linked conclusions were checked against their full text for the specific claims we make, whereas the broader evidence-map citations were characterized at the grouping level from an automated full-text screen, with per-claim audit of the remaining map citations ongoing and released incrementally. None of these bounds touches the core claim, which needs no appendix to verify: a shared robustness label does not imply a comparable result, and the comparability tuple—especially its action coordinate—names the fields that decide. The released framework, adjudicated map, second-coding and adjudication trail, re-analysis worksheets, and reporting checklist make that claim reusable.

## Ethics Statement

**Dual use.** This paper analyzes published robust-RAG attacks and defenses at the level of threat model and evaluation signal, not operational payloads. We introduce no new attack and release no poisoned corpora; the released artifacts are coding schemas, adjudicated evidence-map rows, and analysis worksheets over already-public papers. The aim is to make reliability and security comparisons more transparent and to support safer deployment of retrieval-augmented systems.

**Human annotation.** The reliability and validation studies asked independent expert annotators to code or compare *published papers*; no human-subjects data about the annotators is the object of study. The task carries minimal risk and collects no personal, demographic, health, or otherwise sensitive information about participants. Annotation was voluntary and conducted with informed consent: annotators were told the purpose of the study, the procedure, and the intended research and release use of their anonymized judgments. Public artifacts retain only anonymized participant identifiers and their judgments; annotator identities and credentials are kept private and are not part of the release. Annotators were compensated at or above local hourly norms for expert annotation. The studies were determined to be exempt from full institutional review-board oversight as minimal-risk research on professional task performance with no sensitive personal data; we report this status rather than implying formal approval where none was required.

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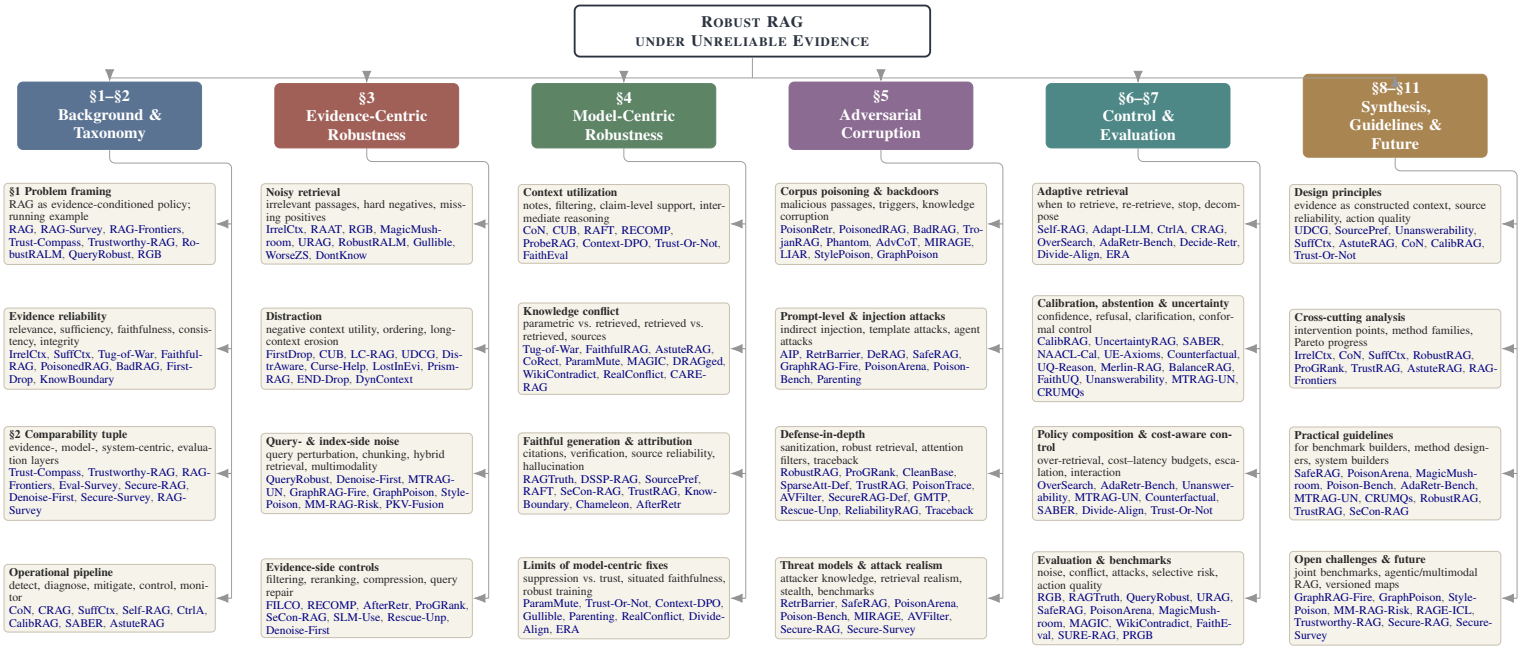



Figure 1: **Outline of this paper.** A top-down view of the paper structure. The root problem is robust RAG under unreliable evidence; the paper then moves from background and taxonomy to evidence-centric, model-centric, adversarial, system-control, evaluation, synthesis, and future-facing guidelines. Each leaf lists representative cited work using compact aliases.

## A Prior Survey Positioning

## B Detailed Background and Taxonomy

## C Evaluation Details

## D Exploratory Map Statistics

| Adjacent survey line | Typical organizing unit | Missing decision field for comparison | What this paper adds |
|---|---|---|---|
| General RAG surveys (Gao et al., 2024) | Retriever/generator architectures, augmentation stages, applications, and broad challenges. | Whether a robustness result tests relevance, sufficiency, consistency, provenance, or integrity; whether the required action is answer, retrieve, filter, verify, or abstain. | A five-coordinate comparison tuple that separates evidence state from module location and final action. |
| Comprehensive RAG/frontier surveys (Sharma, 2025) | Enhancements, optimization modules, and robustness frontiers. | Whether two methods are substitutes or composable pipeline interventions under the same evidence assumption. | A decision rule that labels comparisons as valid, partial, invalid, or under-specified. |
| RAG evaluation surveys (Yu et al., 2024a) | Datasets, metrics, automatic evaluation, and benchmark design. | Whether answer accuracy, citation support, evidence sufficiency, calibration, action quality, utility loss, and cost are measuring the same target. | A survey reading guide that separates metric targets and treats action labels as necessary context for interpreting benchmark claims. |
| Trustworthiness surveys (Zhou et al., 2024; Ni et al., 2025) | Factuality, safety, privacy, fairness, explainability, accountability, and user trust. | Which immediate evidence failure causes the trustworthy-AI risk and which control action should be rewarded. | Evidence-reliability dimensions that connect broad trust axes to concrete RAG failure states. |
| Security-focused RAG surveys (Xu et al., 2026) | Attack surfaces, poisoning, prompt injection, privacy leakage, and defenses. | Corpus write access, query control, retriever visibility, generator access, insertion budget, retrieved malicious evidence, and attack objective. | Threat-model normalization before comparing attack success, prevention, detection, traceback, or recovery. |
| Faithfulness and hallucination-oriented work | Unsupported claims, attribution, citation quality, and factual consistency. | Whether the system had sufficient evidence, used the evidence, cited supporting spans, or should have refused. | A separation between answer correctness, evidence support, citation faithfulness, confidence, and action policy. |

Table 5: Delta relative to adjacent RAG survey lines. The contribution is a comparability framework and reading guide that specifies which decision fields must be aligned before robust-RAG results are interpreted as competing evidence.

| Common survey-level shortcut | Why it is misleading | Our re-interpretation | Design implication |
|---|---|---|---|
| Treating IrrelCtx, RAAT, Chain-of-Note, Sufficient Context, and RGB as one "noise robustness" line. | They test different evidence states and actions: filtering, noise-aware training, generator evidence reading, sufficiency prediction, and benchmark abilities. | The field has decomposed noise robustness rather than ranked one best method. | Benchmarks should cross negative type, rank, support availability, context budget, and answer/abstain/retrieve actions. |
| Grouping RAGTruth-style attribution resources with FaithfulRAG/AstuteRAG conflict methods. | Span-level support, fact-level conflict reasoning, document-level evidence reading, and final answer correctness are different targets. | Faithfulness, conflict, and attribution are complementary diagnostics, not interchangeable metrics. | Evaluation should report claim support, source reliability, and action under conflict, not only citation presence. |
| Comparing conflict methods with poisoning defenses because both involve contradictory retrieved text. | Natural or diagnostic contradiction differs from attacker-created integrity failure with corpus access, insertion budget, and attack objective. | Conflict and poisoning share annotation needs but define different scientific games. | Poisoning defenses require threat-model-indexed evaluation; conflict methods require source reconciliation and disclosure metrics. |
| Ranking prompt-injection defenses, certified defenses, traceback, and poisoning detectors by one attack-success score. | They answer different roles: prevention, demotion, detection, certification, traceback, and recovery. | Security results transfer only inside a capability envelope and an action policy. | Reports should include corpus write access, query control, retriever visibility, budget, objective, clean utility, and defense action. |
| Treating multimodal, graph, table, agentic, and medical RAG as ordinary text RAG applications. | Evidence objects differ: image crops, graph neighborhoods, table cells, memory states, tool calls, and medical sources have different validity conditions. | Modality/scenario RAG is a core extension layer with typed evidence states. | Benchmarks need modality-specific grounding labels and actions before cross-modal results are compared. |

Table 6: Concrete survey deltas. Each row gives an example of a comparison or synthesis shortcut that adjacent surveys may leave underspecified, and the practical consequence of applying our comparability lens.

| Audited source family | Grouping / comparison field inspected | Tuple verdict pattern | Conservative interpretation |
|---|---|---|---|
| General RAG surveys | Retrieval/generation modules and robustness/frontier groupings. | Mostly partial / under-specified. | Useful landscape grouping; not enough to rank robust-RAG mechanisms without action/metric fields. |
| RAG evaluation surveys | Metrics, datasets, automated evaluators, and benchmark families. | Partial. | Evaluation artifacts can be combined, but metric target and action policy remain separate. |
| Trustworthy RAG surveys | Factuality, trust, attribution, safety, and accountability axes. | Partial / under-specified. | Trust axes identify risks but do not always specify immediate evidence state or action. |
| Security RAG surveys | Poisoning, prompt injection, extraction, attacks, and defenses. | Partial / invalid across threat models. | Capability normalization is required before comparing attack/defense numbers. |
| Faithfulness / hallucination lines | Citation support, hallucination detection, and grounded attribution. | Partial. | Support labels are not conflict robustness or action control by themselves. |
| Multimodal/application survey lines | Multimodal, graph, table, domain, and agentic RAG as extensions. | Under-specified for text-RAG transfer. | Passage-level evidence metrics do not automatically transfer to typed evidence objects. |
| Initial audited grouping sample (12 grouping claims) | | 0 valid / 9 partial / 1 invalid / 2 under-specified | We report misread risk, not that prior surveys are wrong. |

Table 7: Prior-survey grouping audit summary. The audit inspects organizing/grouping claims in adjacent survey families and asks whether those groupings are sufficient for direct robust-RAG comparison. It does not claim that the cited surveys make erroneous recommendations; it shows where their broad groupings can leave comparison-critical fields unspecified.

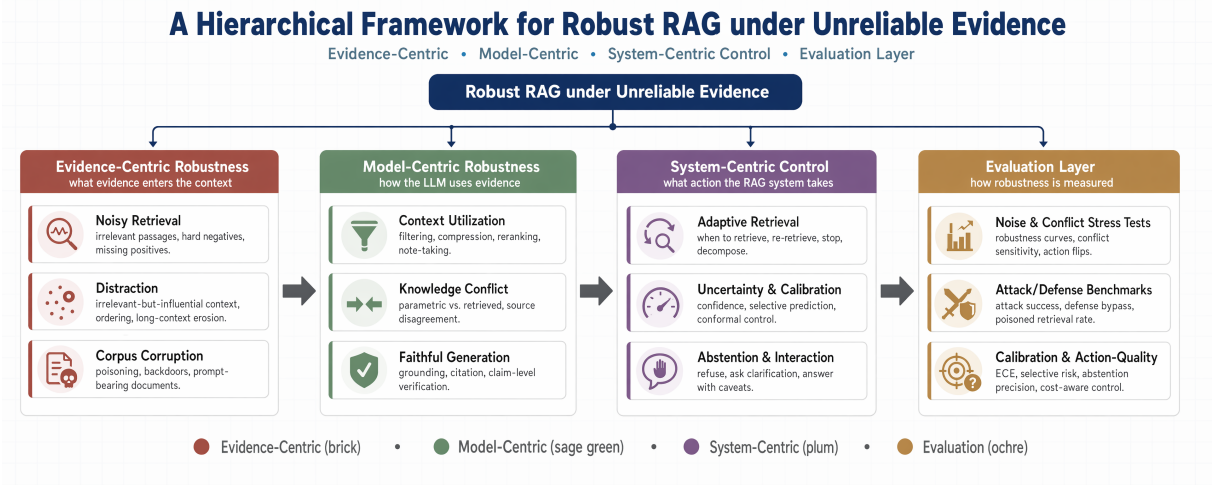



Figure 2: Research landscape map for robust RAG. The figure organizes the literature by where robustness problems and methods sit in the RAG stack: evidence-centric work controls the reliability of retrieved inputs, model-centric work controls how the generator uses evidence, system-centric work studies control decisions, and evaluation work measures robustness and action quality.

| Coordinate | Controlled values | Equivalence / boundary rule | Verdict effect |
|---|---|---|---|
| Failure type | Relevance/noise; sufficiency; consistency/conflict; provenance/freshness; integrity/adversarial corruption; uncertainty/OOD. | Code the evidence property being tested, not the paper's motivation. If a paper only says "robustness" without an evidence property, code the coordinate as unobtainable (U). | Valid comparison requires the same failure type. Same broad area but different failure property gives partial at best. |
| Evidence assumption | Benign retrieval error; benchmark-constructed stressor; naturally occurring conflict; stale or low-quality source; attacker-created evidence; privacy or extraction threat. | Natural contradiction and attacker-created contradiction are different assumptions even if the context text looks similar. For security, record corpus write access, query control, retriever visibility, generator access, insertion budget, and objective when reported. | Different assumption classes usually make a leaderboard comparison invalid; missing attacker capability caps the verdict at under-specified. |
| Intervention point | Corpus/index; retriever; reranker; context assembly/compression; generator prompting or training; verifier; system-control policy; evaluation-only artifact. | Code the first component that changes system behavior. If two methods target different components under the same failure type, compare as pipeline components, not substitutes. | Same failure but different intervention point yields partial unless the paper explicitly studies composition. |
| Metric target | Answer correctness; retrieval recall; evidence support; citation faithfulness; attack success; calibration error; selective risk/coverage; cost/latency; human preference. | Metrics are equivalent only when they measure the same target object. Answer correctness, citation support, and attack success are not interchangeable even on the same dataset. | Different metric families yield partial; conflicting target objects yield invalid for ranking by one scalar. |
| Action policy | Answer; abstain; re-retrieve; filter; verify; expose conflict; ask clarification; escalate; quarantine; recover; unreported. | The action must be explicitly stated or evaluated. It may not be inferred from task setup for a valid verdict. | Unreported action caps the verdict at under-specified. Different explicit actions under the same failure type yield partial. |
| Final verdict | Valid; partial; invalid; under-specified. | Valid = all five coordinates align under the rules above. Partial = same failure but a non-fatal mismatch in intervention, metric, or action. Invalid = assumption or objective defines a different game. Under-specified = at least one required coordinate is U. | The verdict is conservative: under-specification blocks valid comparison even if the papers appear topically similar. |

Table 8: Executable decision procedure for the comparability protocol. The table replaces informal "close enough" matching with coordinate-level equivalence and conservative verdict caps.

| Boundary case | Rule used in this paper | Verdict consequence | Example implication |
|---|---|---|---|
| Noisy retrieval vs. distraction | Retrieval imperfection alone is coded as relevance/noise; harmful influence on generation, confidence, or reasoning is coded as distraction. | Same broad topic but different measured effect is partial. | Random negatives and hard distractors should not be pooled into one score. |
| Natural conflict vs. adversarial corruption | Code by the source of contradiction, not by surface form. Naturally occurring or diagnostic contradiction is conflict; attacker-created contradiction is integrity/adversarial corruption. | Natural-conflict and poisoning results are invalid as a direct ranking unless the threat model is redefined. | Faithfulness or source-preference methods are not poisoning defenses by default. |
| Intervention mismatch | Code the first component that changes system behavior. Different first components under the same failure are complementary unless composition is evaluated. | Same failure with retriever-side and generator-side interventions is partial. | A filter and a note-taking prompt can both address distraction but should not be ranked as substitutes. |
| Metric overlap | Metric targets must match the object being measured. Accuracy, citation support, attack success, calibration, and selective risk are separate families. | Shared dataset but different metric family is partial; mutually incompatible targets are invalid for one-scalar ranking. | A correct answer with unsupported citation should not pass a faithfulness comparison. |
| Unreported action policy | Action policy must be explicit in the method, benchmark label, or reported metric. It cannot be inferred from a QA setup for a valid verdict. | Missing action policy caps the verdict at under-specified. | A system that merely outputs an answer is not automatically an "answer" policy for unreliable evidence. |
| Security capability missing | If an attack or defense omits corpus write access, query control, retriever visibility, generator access, insertion budget, or objective, mark the missing capability as U. | Missing capability caps security comparisons at under-specified; incompatible capabilities make them invalid. | Low attack success under a bounded budget does not transfer to open-corpus prompt injection. |
| Benchmark vs. method | Code benchmark if the reusable artifact is the main contribution; otherwise code the intervention. | Benchmark-method comparison is partial unless the method is evaluated on the benchmark under aligned assumptions. | RGB is benchmark-first; RAAT is training-first. |

Table 9: Hard boundary rules for applying the protocol. These rules address cases where topical similarity is insufficient for a valid comparison.

| Stage | Question | Typical signals | Representative actions |
|---|---|---|---|
| Detect | Is the current evidence unreliable? | Low retrieval confidence, high contradiction, low sufficient-context score, suspicious document patterns | Flag query/evidence, route to stronger checks, log risk state |
| Diagnose | What kind of unreliability is present? | Irrelevance, distraction, conflict, poisoning, ambiguity, unanswerability, stale evidence | Assign failure type; separate benign noise from adversarial risk |
| Mitigate | Can the evidence be repaired before answering? | Alternative retrieval results, filter scores, reranker utility, provenance, source agreement | Re-rank, filter, compress, re-query, retrieve from trusted sources, remove malicious passages |
| Control | What should the system do now? | Calibrated confidence, selective risk, cost budget, user risk tolerance | Answer, abstain, ask clarification, expose conflict, escalate to human review |
| Monitor | Did the system fail or get attacked? | User feedback, traceback signals, poisoned-document clusters, drift over time | Update index, quarantine sources, revise prompts, retrain filters, update audit logs |

Table 10: An operational pipeline for robust RAG. The pipeline turns the paper's taxonomy into actionable system design steps.

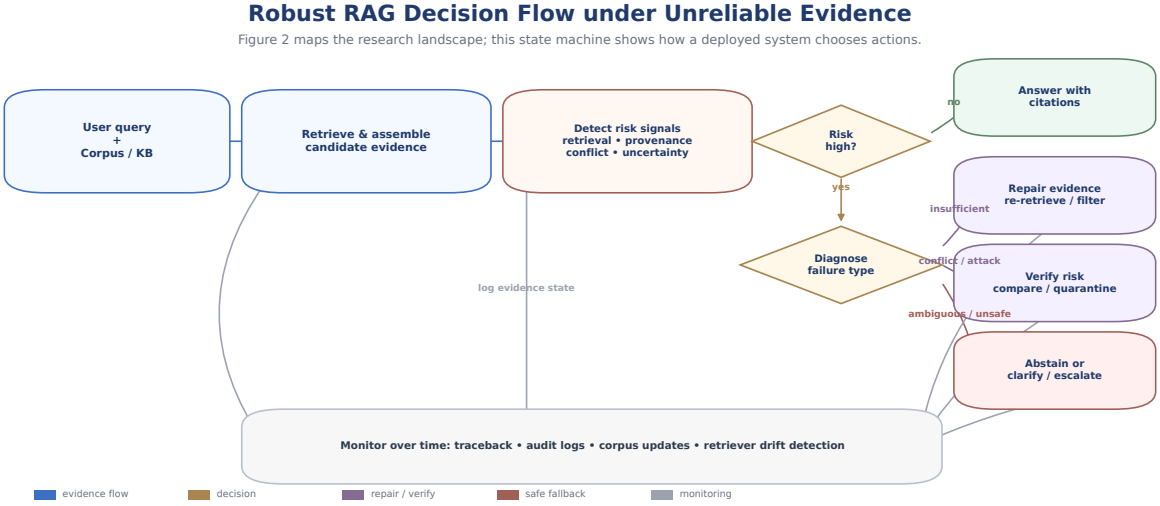



Figure 3: Operational decision flow for robust RAG. Unlike the taxonomy in Figure 2, this state-machine view describes the branching actions a deployed system can take when unreliable evidence is encountered: answer, repair evidence, verify risk, abstain, clarify, escalate, or monitor.

| Theme | Representative work | Core idea | What it reveals for robust RAG |
|---|---|---|---|
| Irrelevant context | IrrelCtx | Analyze and mitigate irrelevant retrieved context with filtering and robust training | Even answerable questions degrade when irrelevant passages enter the context; robustness is not equivalent to retrieval recall. |
| Retrieval-noise training | RAAT | Categorize retrieval noise and train with adaptive adversarial objectives | Noise types should be separated: irrelevant, relevant-but-insufficient, and counterfactual evidence have different effects. |
| Generator note-taking | CoN | Generate per-passage notes before answering | Robust generation benefits from intermediate relevance judgments rather than direct answer generation. |
| Fine-grained noise benchmark | MagicMushroom | Configurable benchmark for distracting, low-quality, inconsequential, and irrelevant noise | "Noise" is not monolithic; evaluation should vary type, amount, and placement of bad evidence. |
| Nonlinear distraction | FirstDrop | Study disproportionate effect of small amounts of misleading information | A few strong distractors can dominate long-context reasoning, so average noise rate is insufficient. |
| Long-context RAG | LC-RAG | Analyze long-input RAG and hard negatives | More retrieved context can hurt unless ordering, filtering, or training adapts to long-context failure modes. |
| Sentence-level filtering | FILCO | Filter retrieved context before generation | Filtering can be more fine-grained than document reranking and can remove local distractors inside otherwise useful passages. |
| Retrieval evaluation | UDCG | Utility-aware retrieval metric for LLMs | RAG retrieval should be evaluated by downstream utility, including negative utility from distracting passages. |

Table 11: Representative evidence-centric work on noisy retrieval and distraction.

| Problem | Representative work | Mechanism | Limitation or open issue |
|---|---|---|---|
| Parametric–retrieved conflict | Tug-of-War | Construct and analyze conflicts between internal and external knowledge | Conflict behavior depends on model, prompt, evidence coherence, and task framing. |
| Fact-level conflict modeling | FaithfulRAG | Extract parametric self-facts, align them with retrieved facts, and reason before answering | Requires reliable fact extraction and alignment; complex claims remain difficult. |
| Source-aware consolidation | AstuteRAG | Elicit internal knowledge and consolidate it with retrieved sources using reliability comparison | Training-free but still depends on LLM self-assessment and source-quality cues. |
| Hidden-state rectification | CoRect | Locate layers where parametric suppression occurs and contrast contextual/non-contextual logits | Promising for model-centric control, but model internals and transferability remain open. |
| Context utilization benchmarks | CUB | Test gold, conflicting, and irrelevant contexts across models | Benchmarks reveal whether a model uses context, but not always why a model chooses one source. |
| Real-document conflict | RealConflict | Study behavior under context-memory conflicts with real documents | Realistic evidence can be messier than synthetic conflicts and harder to label. |
| Source preference | SourcePref | Analyze which sources win under conflict | Source preference may encode bias, recency effects, or authority priors. |

Table 12: Representative model-centric work on knowledge conflict and contextual faithfulness.

| Attack/defense surface | Representative work | Threat or defense | Implication |
|---|---|---|---|
| Retriever poisoning | PoisonRetr | Inject adversarial passages to manipulate dense retrievers | Retrieval security can fail before generation begins. |
| RAG knowledge corruption | PoisonedRAG | Craft malicious passages that steer generated answers | End-to-end RAG needs corpus integrity and generation safeguards. |
| Backdoor attacks | TrojanRAG; Phantom | Triggered retrieval/generation behavior via poisoned contexts | RAG can act as a backdoor carrier without modifying model weights. |
| Prompt template attacks | AIP | Weaponize shared instructional prompts to steer retrieval and generation | Prompt templates are part of the attack surface, not just implementation details. |
| Indirect prompt injection | RetrBarrier | Optimize trigger fragments so malicious instructions are retrieved in the wild | Real attacks must overcome the retrieval barrier, not merely place malicious text in a corpus. |
| Certified defense | RobustRAG | Isolate-and-aggregate answers for robustness certificates | Certification gives stronger guarantees but typically assumes specific attack budgets. |
| Reranking defense | ProGRank | Use probe-gradient instability to demote poisoned passages | Defense can be inserted retriever-side without retraining the LLM. |
| Attention defense | AVFilter | Detect poisoning via normalized attention variance | Internal attention signals may expose poisoned evidence, but adaptivity remains a concern. |

Table 13: Representative adversarial corruption attacks and defenses.

| Threat class | Corpus | Query | Retr. | Gen. | Objective | Boundary |
|---|---|---|---|---|---|---|
| Dense-retriever poisoning (Zhong et al., 2023) | write | train set | white/gray | no | retrieve bad passages | Not always end-to-end. |
| RAG knowledge corruption (Zou et al., 2025; Xue et al., 2024) | write | target topic | black/gray | black | answer steering | Stealth/dynamics vary. |
| Backdoor-style RAG (Cheng et al., 2024a; Chaudhari et al., 2024) | write | trigger | gray | black | latent behavior | Normal queries may pass. |
| Indirect prompt injection (Chaturvedi et al., 2025; Chang et al., 2026) | write/share | natural | black | black | instruction hijack | Must overcome retrieval. |
| Opinion/fairness manipulation (Chen et al., 2025i; Bagwe et al., 2025) | write | group/topic | black/gray | black | bias steering | QA metrics miss social harm. |
| Defense setting (Xiang et al., 2024; Yin et al., 2026; Choudhary et al., 2025; Zhang et al., 2025b) | bounded | varies | assumed | black/gray | detect/demote/trace | Depends on budget/visibility. |

Table 14: Attacker-capability matrix for adversarial RAG. The same "attack success rate" can mean different things depending on corpus write access, query control, retriever visibility, generator access, and objective.

| Comparison | Why it looks comparable | Hidden mismatch | Metric pitfall | Normalized reading |
|---|---|---|---|---|
| PoisonedRAG vs. dense-retriever poisoning | Both insert malicious evidence into a retrieval corpus. | One evaluates end-to-end answer steering; the other may evaluate retriever ranking before generation. | Retrieval success and answer attack success are not the same target. | Compare only after recording whether poisoned evidence is retrieved and whether the generator changes the final answer. |
| Certified RAG defense vs. prompt-injection defense | Both are RAG defenses against corrupted retrieved content. | Certification assumes a bounded number of corrupted passages; prompt injection exploits instruction following rather than only false facts. | A certificate under bounded corruption may not imply instruction-hijack resistance. | Treat certification, instruction isolation, and document sanitization as complementary defenses. |
| Traceback vs. prevention | Both reduce harm after poisoning. | Traceback identifies influential poisoned documents after failure; prevention attempts to block harmful output before response. | Lower post-hoc traceback error does not equal lower user-facing attack success. | Report prevention, detection, demotion, traceback, and recovery as separate outcome variables. |
| Backdoor trigger vs. natural query attack | Both can produce targeted malicious behavior. | Trigger attacks may rely on rare phrases; natural-query attacks must survive ordinary retrieval and ranking. | Attack success under attacker-crafted triggers overstates risk under natural user queries. | Require query-control and trigger-frequency fields before comparing attack strength. |

Table 15: Concrete examples of threat-model normalization. These cases show how reported attack or defense success can reverse when attacker access, retrieval realism, and evaluation target differ.

| Control/evaluation question | Representative work | Signal or decision | Why it matters |
|---|---|---|---|
| When to retrieve | Self-RAG; Adapt-LLM; CtrlA | Reflection tokens, special retrieval tokens, or representation-based confidence/honesty directions | Retrieval is an action with costs and risks, not a fixed preprocessing step. |
| When retrieval is weak | CRAG; SuffCtx | Retrieval evaluator or sufficient-context signal | A system should repair or abstain rather than answer from insufficient evidence. |
| Calibration | CalibRAG; UncertaintyRAG; NAACL-Cal | Decision calibration, span uncertainty, verbal confidence calibration | Robustness requires confidence to reflect both model and evidence uncertainty. |
| Unanswerability | Unanswerability; MTRAG-UN | Unanswerable and underspecified RAG benchmarks | Real users often ask questions the corpus cannot answer; refusal and clarification need direct evaluation. |
| RAG capabilities | RGB | Noise robustness, negative rejection, integration, counterfactual robustness | Multi-axis evaluation exposes failures hidden by answer accuracy. |
| Hallucination detection | RAGTruth | Word-level hallucination annotations for RAG outputs | Fine-grained labels support detection and attribution beyond answer-level scoring. |
| Query robustness | QueryRobust | Perturb query wording and measure pipeline sensitivity | Robustness must include the query-to-retrieval interface. |

Table 16: Representative work on system-centric control and evaluation.

| Lineage | Classical question | RAG-specific version | Why it matters for robust RAG |
|---|---|---|---|
| Selective prediction | When should a model abstain to control risk? | Abstain when retrieved evidence is insufficient, conflicting, stale, or corrupted. | Selective risk must condition on evidence state, not only answer confidence. |
| Calibration | Does confidence match correctness? | Does confidence match answer correctness, evidence sufficiency, citation support, and retrieval usefulness? | A calibrated answer probability can still hide unreliable evidence. |
| Active clarification | When should the system ask a user for more information? | Ask clarification before retrieval when the query is ambiguous or underspecified. | Clarification can be the correct action even when retrieval could produce a plausible answer. |
| Cost-sensitive retrieval | When is additional computation worth it? | Decide whether to retrieve, re-retrieve, verify, rerank, or escalate under latency/cost budgets. | Robustness gains should be reported with action cost and coverage, not only accuracy. |
| Decision-theoretic control | Which action minimizes expected risk? | Choose among answer, abstain, expose conflict, cite uncertainty, retrieve again, or quarantine evidence. | Action quality is the bridge between evidence reliability estimates and user-facing behavior. |

Table 17: Action-control lineage. The framework imports ideas from selective prediction, calibration, clarification, and cost-sensitive decision-making, but makes retrieved evidence state the object being controlled.

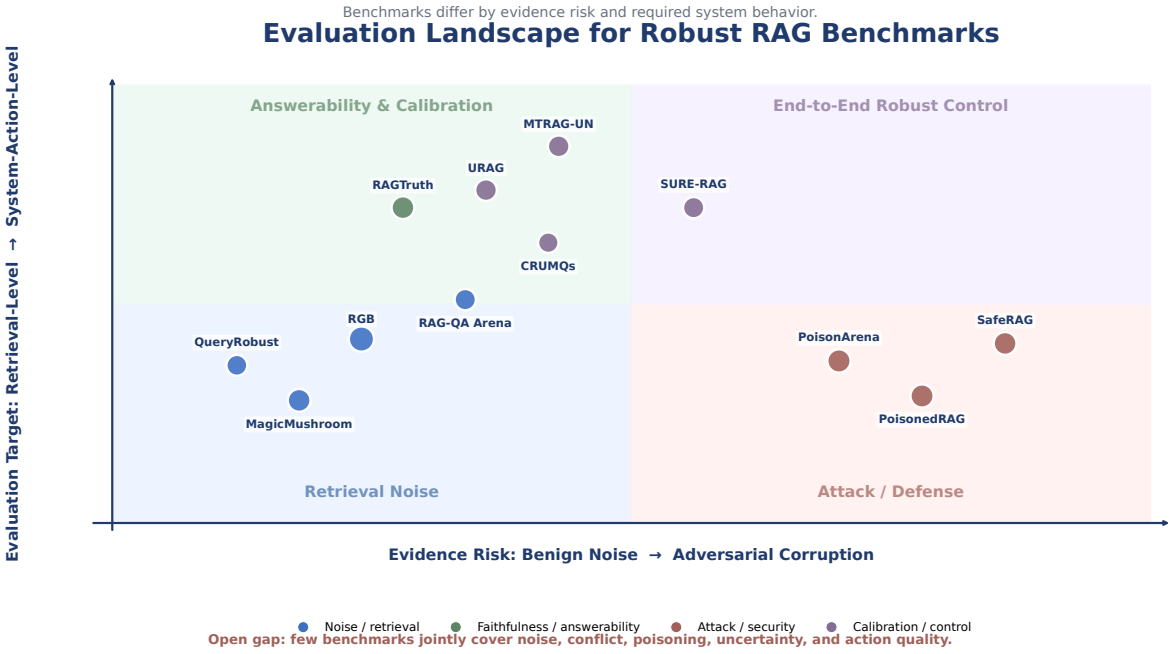



Figure 4: Evaluation landscape for robust RAG benchmarks. Existing benchmarks differ in the type of evidence risk they simulate and whether they evaluate retrieval-level behavior, generation-level behavior, or system-action-level robustness.

| Benchmark | Risk focus | Task / data | Perturbation or threat | Metrics | Boundary |
|---|---|---|---|---|---|
| RGB (Chen et al., 2024a) | noise, rejection, integration, counterfactuality | English/Chinese QA | noisy passages, false evidence, negative evidence | accuracy, ability-specific scores | Broad diagnostic benchmark but not adaptive attack-oriented. |
| RAGTruth (Niu et al., 2024) | hallucination / faithfulness | QA, data-to-text, summarization | naturally generated unsupported spans | word-level hallucination labels | Strong annotation granularity; less focused on retrieval-side threat models. |
| Sufficient Context (Joren et al., 2025) | answerability | QA with context sufficiency labels | insufficient retrieved evidence | selective generation / abstention | Captures answerability but not adversarial corruption. |
| MagicMushroom (Zhang et al., 2025g) | retrieval noise | configurable RAG stress tests | distracting, low-quality, inconsequential, irrelevant noise | robustness under noise types | Fine-grained noise; limited conflict/security coverage. |
| URAG (Nguyen et al., 2026) | uncertainty | RAG as MCQA / conformal prediction | irrelevant/noisy retrieval and uncertainty sets | coverage, set size, calibration | Good for uncertainty; less direct for free-form generation. |
| QueryRobust (Percin et al., 2025) | query robustness | QA with query perturbations | wording variation at query–retrieval interface | retrieval and end-task degradation | Focuses on query perturbation, not corpus corruption. |
| PoisonArena (Chen et al., 2025e) | adversarial poisoning | RAG attack competition setting | competing poisoning attacks | attack success, retrieval effects | Strong security focus; benign uncertainty is secondary. |
| SafeRAG (Liang et al., 2025b) | RAG security | security benchmark for RAG | poisoning / malicious retrieval contexts | attack and defense metrics | Security-oriented; not a full evidence reliability suite. |
| MTRAG-UN (Rosenthal et al., 2026) | multi-turn unanswerability | multi-turn RAG conversations | unanswerable, underspecified, non-standalone questions | abstention / clarification quality | Covers interaction; less focused on adversarial evidence. |
| CRUMQs (Liu et al., 2025a) | unanswerable multi-hop RAG | realistic multi-hop queries | unanswerable and uncheatable questions | answerability and robustness | Useful for abstention; not a broad attack/noise benchmark. |

Table 18: Benchmark-level comparison for robust RAG evaluation. The table makes explicit what each benchmark perturbs, what it measures, and which failure modes remain outside its scope.

| Benchmark component | Required annotation | Metric suite | Why it is needed |
|---|---|---|---|
| Evidence reliability | relevance, sufficiency, freshness, provenance, support span | retrieval recall, sufficient-context accuracy, support precision | Distinguishes merely related passages from answer-supporting evidence. |
| Distraction and conflict | distractor type, conflicting claim pair, source agreement | answer flip rate, conflict sensitivity, context-utilization score | Reveals whether models ignore harmful context or silently pick a source. |
| Adversarial corruption | attacker access, insertion budget, trigger type, poisoned document ID | attack success, poisoned retrieval rate, stealth, defense bypass | Makes security results comparable across threat models. |
| Unanswerability and ambiguity | answerability label, ambiguity type, clarification target | selective risk, abstention precision/recall, clarification accuracy | Evaluates robust action selection, not just answer generation. |
| Cost and system action | number of retrievals, verifier calls, latency, final action trace | cost-adjusted accuracy, action utility, coverage–risk curve | Prevents methods from hiding robustness gains behind unbounded computation. |

Table 19: A proposed benchmark blueprint for robust RAG under unreliable evidence. This is a research agenda rather than a validated leaderboard; existing benchmarks cover parts of the schema.

| Example query | Retrieved evidence state | Correct action | Wrong action | Metric consequence |
|---|---|---|---|---|
| "What is the refund deadline in policy version 3?" | Gold policy passage is present, but a high-ranked older policy gives a different deadline. | Compare provenance/date, prefer the current policy, cite the supporting span. | Answer from the older top-ranked passage. | Answer accuracy alone may mark one string right/wrong; action-aware scoring credits source freshness and conflict handling. |
| "Which paper introduced method X's ablation result?" | Retrieved documents are topically related but none contains the requested ablation. | Detect insufficiency and re-retrieve or abstain with missing-evidence explanation. | Guess from a related abstract. | Selective risk and sufficiency detection should be rewarded; forced-answer accuracy should not. |
| "Summarize whether product Y is safe for use case Z." | Retrieved reviews contain plausible positive statements and a contradictory safety notice from an authoritative source. | Expose conflict, prioritize authoritative safety evidence, or escalate. | Produce a balanced answer that hides the conflict. | Faithfulness to a retrieved passage is not enough; metric must score conflict disclosure and source authority. |
| "What does the linked document instruct the assistant to do?" | A retrieved document contains task-irrelevant instructions mixed with factual content. | Treat instructions as untrusted evidence, filter or quarantine, then answer only from factual support. | Follow the retrieved instruction. | Attack success must be separated from poisoned retrieval and from final answer utility. |
| "Should the system answer now or search again?" | Current evidence is relevant but incomplete; a second retrieval is available at latency cost. | Choose re-retrieval if expected utility exceeds cost, otherwise abstain. | Answer immediately with overconfident unsupported text. | Calibration error alone is insufficient; score action utility, coverage, latency, and evidence sufficiency. |

Table 20: Illustrative action-labeled evaluation blueprint. Rows are examples of the annotation schema implied by the re-analyses, not a released benchmark or empirical leaderboard.

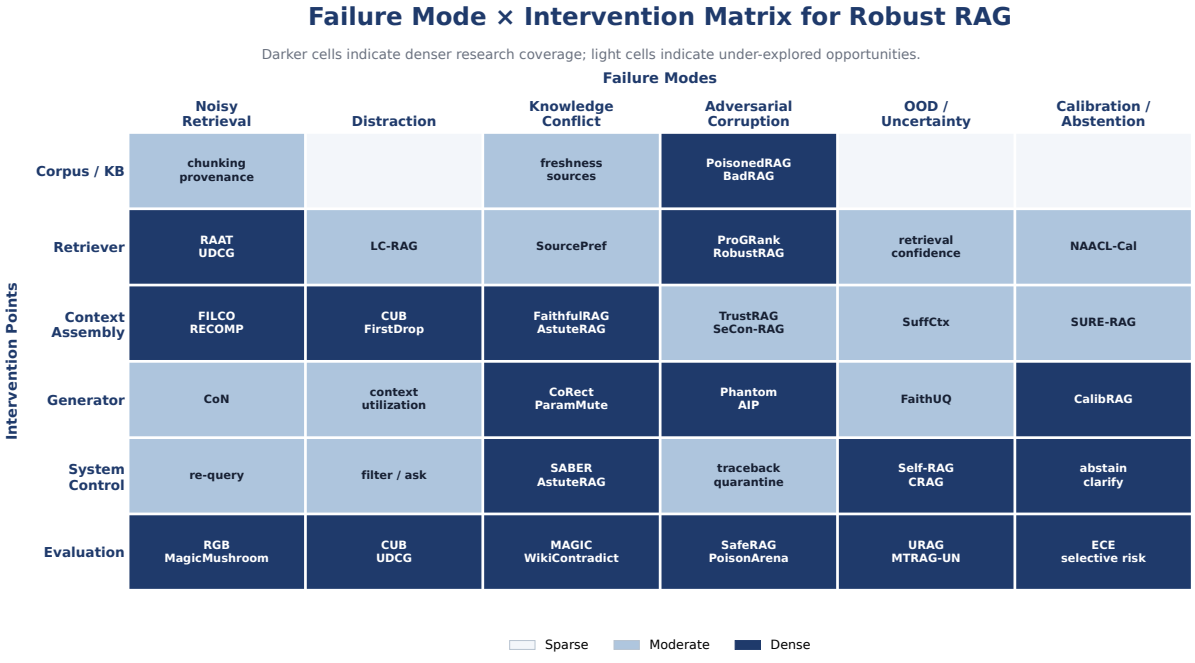



Figure 5: Failure mode × intervention matrix for robust RAG. Rows indicate where a method intervenes in the pipeline, while columns indicate the failure mode being addressed. Darker cells denote denser research coverage and lighter cells indicate under-explored opportunities.

| Policy | Ans. | Act. | Joint | Unsafe |
|---|---|---|---|---|
| Answer-only | 0.17 | 0.17 | 0.17 | 0.83 |
| Abstain-on-missing | 0.67 | 0.33 | 0.33 | 0.17 |
| State-label heuristic | 1.00 | 1.00 | 1.00 | 0.00 |

Table 21: Minimal action-labeled pilot over 60 diagnostic items and three simple policies. The state-label heuristic uses the gold evidence-state label and is an upper-bound sanity check, not a deployed model. Unsafe is the rate of answering when the gold action is to abstain, quarantine, clarify, expose conflict, or re-retrieve.

| Axis | Typical variation across papers | Why conclusions fail to transfer | Minimum reporting requirement |
|---|---|---|---|
| Retriever/generator setting | BM25 vs dense vs hybrid retrieval; small vs frontier generators; frozen vs fine-tuned models | A defense may work by compensating for one retriever or one model scale rather than solving the failure mode. | Retriever, index, top-$k$, reranker, generator, prompt, decoding, and context length. |
| Attack budget and access | Poisoned-document count, corpus write access, query control, retriever visibility, generator access | Attack success rates are incomparable when one attacker edits a corpus and another only controls the query. | Capability matrix, insertion budget, target set, stealth objective, and whether malicious evidence is retrieved. |
| Metric target | EM/F1, faithfulness, citation support, attack success, calibration error, selective risk, cost | Improving answer accuracy can worsen citation faithfulness; improving abstention can reduce coverage; reducing attacks can lower utility. | Report utility, evidence support, risk, calibration, cost, and security as a Pareto surface. |
| Statistical and reproducibility evidence | Single seed/model vs multiple seeds/models; code/data availability varies | Small prompt or model changes can reverse robustness rankings, especially for LLM-as-judge or attack benchmarks. | Seeds, confidence intervals or bootstrap tests where feasible, released prompts, data splits, and code or exact API settings. |
| Long-context baseline | RAG-only comparisons vs long-context LLMs with the same documents | A robustness gain may reflect retrieval compression rather than evidence reliability control. | Include a long-context or full-context baseline when context length permits, and report latency/cost. |
| Attribution vs answer accuracy | Some tasks score only final answer; others require evidence-supported citations | A model can answer correctly for the wrong reason, cite irrelevant evidence, or ignore retrieved support. | Claim-level support labels, citation precision/recall, and answer correctness reported separately. |

Table 22: Evaluation-assumption matrix for robust RAG. The table summarizes why benchmark results often cannot be compared without aligning system setup, threat model, metric target, reproducibility evidence, and long-context baselines.

| Work | Failure | Task / setup | Key assumption | Metric | Boundary |
|---|---|---|---|---|---|
| IrrelCtx | irrelevant context | Open-domain QA | Irrelevance can be filtered or learned against | Robust accuracy | No adversarial/conflict setting. |
| RAAT | retrieval noise | Noisy QA training | Noise-aware training improves generation | F1/EM | Needs deployment-like noise. |
| RGB | multi-ability RAG | Bilingual benchmark | RAG reliability decomposes into abilities | Ability accuracy | Not adversarially adaptive. |
| FaithfulRAG | conflict | Fact-level conflict | Conflicts can be localized and reasoned over | Faithfulness | Needs reliable fact extraction. |
| AstuteRAG | imperfect retrieval | Training-free consolidation | Internal/external knowledge can be reliability-compared | Robust QA | Depends on LLM self-assessment. |
| PoisonedRAG | poisoning | Injected passages | Few crafted passages can dominate retrieval | Attack success | Targeted corruption focus. |
| RobustRAG | defense | Bounded corruption | Isolate-and-aggregate can certify robustness | Certificate, accuracy | Assumes attack budget. |
| Self-RAG / CtrlA | adaptive retrieval | Retrieve-or-answer | Model state can trigger retrieval | Trigger quality | Calibration can shift. |
| CalibRAG / URAG | calibration | Decision/MCQA | Confidence should reflect evidence uncertainty | ECE, coverage | Does not repair evidence. |
| PoisonArena / SafeRAG | security eval. | Poisoning benchmark | Threat models must be explicit | ASR, bypass | Still isolates attacks. |

Table 23: Analytical matrix of representative robust RAG work. Compared with a citation list, the matrix highlights each work's task setup, central assumption, metric focus, and boundary of applicability.

| Case | Released rows / papers | Audited comparison fields | Metric or failure conflict | Synthesis / artifact boundary |
|---|---|---|---|---|
| Noise / distraction robustness | 5 rows: IrrelCtx, RAAT, Chain-of-Note, Sufficient Context, RGB | Assumption, QA dataset or benchmark slice, generator / reader setup, negative type, rank and gold-evidence availability. | EM/F1, rejection, selective risk, and ability-specific benchmark scores answer different questions. | Treat filtering, noise-aware training, generator reading, sufficiency prediction, and benchmark diagnosis as different mechanisms; artifact status is recorded per paper in the released worksheet. |
| Faithfulness, conflict, and attribution | 5 rows: RAGTruth, FaithfulRAG, AstuteRAG, Tug-of-War, Chain-of-Note | Unit of support, source-conflict assumption, attribution granularity, model-side reasoning setup, artifact availability. | Span-level support, fact-level conflict resolution, source preference, and final answer accuracy can disagree. | Citation faithfulness should not be collapsed into conflict robustness; the worksheet records where artifact evidence is dataset-level, code-level, or paper-only. |
| Adversarial corruption and RAG security | 6 rows: PoisonedRAG, corpus poisoning, BadRAG, certified defense, TrojanRAG, traceback | Corpus write access, insertion budget, retriever visibility, query control, generator access, stealth, retrieved malicious evidence, defense timing. | Attack success, certified robust accuracy, detection, traceback, and clean utility are not one defense score. | Compare only inside a capability envelope; prevention, demotion, detection, traceback, recovery, and certification remain separate outcomes. |
| Calibration, abstention, and adaptive control | 5 rows: Sufficient Context, Self-RAG, CRAG, CalibRAG, uncertainty RAG | Confidence target, evidence sufficiency labels, retrieval trigger, action set, cost budget, model/policy setup. | ECE, selective risk, coverage, retrieval-trigger quality, and answer utility can move in different directions. | Separate belief estimation from action selection; report coverage, cost, and final action alongside correctness. |
| Evaluation and benchmark comparability | 5 rows: RGB, RAGTruth, Sufficient Context, query robustness, RAG security benchmark | Evidence-state labels, perturbation source, model outputs scored, threat model, benchmark artifact availability. | Harder scores may reflect retrieval instability, unsupported spans, unanswerability, or adversarial steering. | Evaluation resources are diagnostic slices, not a universal robustness scale; benchmark comparisons require aligned evidence state and action policy. |

Table 24: Deeper case-study synthesis. The released 26-row worksheet compares assumptions, datasets or benchmark settings, model/evidence setup, metrics, failure modes, artifact availability, and comparison implications for the same five C1–C5 cases.

| Case | Naive reading | Normalization coordinate that changes it | Revised conclusion |
|---|---|---|---|
| IrrelCtx (Yoran et al., 2024) vs. RAAT (Fang et al., 2024) | Both are "noise-robust RAG" methods. | IrrelCtx targets naturally retrieved irrelevant context and filtering/back-off; RAAT trains on constructed relevant/irrelevant/counterfactual mixtures. | They are complementary context-filtering vs. noise-training regimes, not one leaderboard. |
| IrrelCtx (Yoran et al., 2024) vs. Chain-of-Note (Yu et al., 2023) | Both reduce distractor harm. | Intervention point differs: context filtering/fine-tuning vs. generator-side document reading. | A valid comparison must hold context state fixed and report whether the action is filter, answer, or unknown. |
| Sufficient Context (Joren et al., 2025) vs. Self-RAG (Asai et al., 2024a) | Both decide when retrieval is useful. | Evidence-sufficiency labeling vs. retrieve/generate/critique action policy. | Sufficiency prediction is an evidence-state signal; adaptive retrieval is a policy over actions and costs. |
| RAGTruth (Niu et al., 2024) vs. FaithfulRAG/AstuteRAG (Zhang et al., 2025f; Wang et al., 2025a) | All are faithfulness/conflict papers. | Unit of support differs: generated span support vs. fact/source conflict resolution. | Detector F1 and conflict-resolution accuracy should not be pooled as one faithfulness score. |
| Tug-of-War (Jin et al., 2024b) vs. ParamMute (Huang et al., 2025a) | Both study parametric-vs-contextual knowledge. | Source-preference analysis vs. parametric-knowledge suppression/intervention. | The conclusion shifts from "trust context or model" to reporting source reliability and update policy. |
| Poisoning corpora (Zhong et al., 2023) vs. PoisonedRAG (Zou et al., 2025) | Both are RAG poisoning attacks. | Corpus-write access, query control, retriever visibility, and target objective differ. | Attack success rates transfer only inside a shared capability envelope. |
| PoisonedRAG (Zou et al., 2025) vs. CertifiedRAG (Xiang et al., 2024) | Attack and defense appear directly comparable. | Defense guarantee assumes bounded retrieved corruption and an aggregation/abstention policy. | Certification is meaningful only under the stated budget; it is not a universal defense to open-web insertion. |
| AgentPoison (Chen et al., 2024e) vs. document poisoning (Zou et al., 2025) | Both corrupt retrieved evidence. | Persistent memory/tool state vs. static corpus passages. | Agentic poisoning needs memory-update and tool-call action labels, not only answer steering. |
| Spill-Beans/privacy (Unknown, 2025) vs. steering attacks (Zou et al., 2025) | Both are security failures. | Objective differs: data extraction or privacy leakage vs. answer manipulation. | Privacy, extraction, steering, traceback, and recovery require separate outcome columns. |
| RAGAS/ARES (Es et al., 2024; Saad-Falcon et al., 2024) vs. RAGTruth (Niu et al., 2024) | All evaluate grounded RAG. | LLM-as-judge metric toolkit vs. human-annotated hallucination/support resource. | Automated evaluation and corpus labels are different evidence artifacts; agreement with one does not validate the other. |
| RGB (Chen et al., 2024a) vs. Sufficient Context (Joren et al., 2025) | Both benchmark unreliable evidence. | Diagnostic ability axes vs. answerability/sufficiency labels and selective risk. | A benchmark score is interpretable only after stating the evidence state and required action. |
| Lost-in-the-Middle (Liu et al., 2024) vs. long-context retrieval | More context seems to solve retrieval misses. | Evidence availability differs from evidence selection and citation. | Full-context access can still fail if the model cannot select or ground the supporting span. |
| ColPali/Table/Graph RAG (Faysse et al., 2025; Chen et al., 2024c; He et al., 2024) vs. passage RAG | Extension papers look like new application areas. | Evidence object changes: page image, table cell/schema, graph neighborhood, or tool memory. | They are retained as extension-layer stress tests of typed evidence states, not as broad co-equal survey scope. |

Table 25: Concrete conclusions that change after five-coordinate normalization. Each row is grounded in released case-study or Map-B records and shows why a common citation-list comparison becomes invalid, partial, or more specific once failure, assumption, intervention, metric, and action are aligned.

| Naive comparison | Tuple summary (F/A/I/M/Act) | Breaking coordinate(s) | Verdict | What a valid comparison would require | Changed interpretation |
|---|---|---|---|---|---|
| IrrelCtx vs. RAAT vs. RGB as "noise robustness" | Shared relevance/noise; benign or benchmark stressors; generator sensitivity vs. training vs. benchmark; accuracy/ability metrics; action mostly unreported. | Intervention, metric, action. | Partial / under-specified | Fix negative type, rank, gold-evidence availability, generator, metric family, and explicit answer/abstain policy. | They map different operating regions: generator selectivity, noise-aware training, and diagnostic benchmark design. |
| Context filtering vs. Chain-of-Note as distraction mitigation | Shared distraction; benign context noise; context assembly vs. generator reasoning; answer robustness; answer action implicit. | Intervention and action. | Partial | Evaluate both on the same retrieved contexts with the same generator, answer policy, and evidence-support metric. | A filter and a reasoning prompt can compose; ranking them as substitutes hides pipeline position. |
| Sufficient Context vs. Self-RAG as control | Sufficiency vs. retrieval-need uncertainty; benign missing evidence; sufficiency predictor vs. learned retrieve/critique policy; selective risk vs. utility/cost; abstain vs. retrieve/answer. | Failure, intervention, metric, action. | Partial | Hold evidence-sufficiency labels fixed and score the same action set: answer, abstain, retrieve, or critique. | One estimates whether evidence is enough; the other learns a broader control policy. |
| Calibration methods vs. abstention methods | Uncertainty; benign or OOD setting; confidence estimator vs. action policy; calibration error vs. selective risk; action often absent vs. abstain. | Metric and action. | Partial / under-specified | Define whether confidence targets answer correctness, evidence sufficiency, retrieval usefulness, or action utility, then report coverage and risk. | Better calibration is not automatically better abstention. |
| Knowledge-conflict methods vs. PoisonedRAG | Consistency conflict vs. integrity corruption; natural/diagnostic contradiction vs. attacker-created evidence; source reasoning vs. attack/defense; reconciliation vs. attack success; answer/expose conflict vs. resist steering. | Assumption, metric, action. | Invalid | Recast both under the same threat class and objective, or evaluate conflict detection separately from adversarial steering. | Conflict annotations can help both areas, but metrics and threat games differ. |
| Corpus poisoning vs. prompt injection | Integrity corruption; corpus insertion vs. instruction-bearing retrieved text; retrieval/corpus defense vs. instruction-following defense; attack success; prevention/demotion/refusal. | Assumption and objective. | Invalid | Specify whether the attacker controls documents, user query, prompt instructions, or model memory, and evaluate the same objective. | Both use malicious text, but factual steering and instruction hijacking are different security games. |
| PoisonedRAG vs. certified defenses | Shared integrity corruption; empirical corpus injection vs. bounded corruption budget; attack benchmark vs. certified aggregation; attack success vs. certified robust accuracy; prevention implicit. | Assumption, metric, action. | Under-specified / invalid for ranking | Align corruption budget, corpus access, retrieval success, generator access, utility loss, and defense action. | A certificate under one budget does not imply robustness to open-ended poisoning or prompt-bearing documents. |
| RAGTruth vs. answer-accuracy benchmarks | Generated-output evaluation; natural unsupported claims vs. task correctness; evaluation artifact; citation/support vs. exact answer correctness; action unreported. | Metric and action. | Partial | Score the same system outputs with both answer correctness and claim-level support, and define how unsupported correct answers are handled. | Correct answers can still be unreliable if unsupported or falsely cited. |
| Long-context RAG vs. retrieval-noise benchmarks | Evidence selection under many tokens vs. top-$k$ retrieval noise; context-length manipulation vs. retrieval perturbation; accuracy/utility; action usually answer-only. | Failure construction, intervention, action. | Partial | Match evidence position, distractor density, retrieval requirement, context budget, and action labels. | Long context changes the bottleneck from retrieval recall to evidence selection and attention. |
| Two poisoning defenses with missing utility reporting | Shared integrity corruption may hold; threat capabilities may be incomplete; detection/demotion/prevention may differ; attack success reported without clean-utility loss; action implicit. | Missing capability, metric, action. | Under-specified | Report attacker budget, retriever visibility, query control, clean accuracy, latency, and whether the defense prevents, detects, demotes, traces, or recovers. | Low attack success is not portable without utility and capability normalization. |

Table 26: Normalized re-analysis worksheet. Each row records the five-coordinate tuple, the coordinate that blocks direct comparison, the conservative verdict, and the information required to make a valid comparison. The table is a reproducible coding artifact, not a re-run of the original systems.

| Step | Rule | Artifact | Why it matters |
|---|---|---|---|
| Sampling | Select paper groups that appear in adjacent-survey organizing units or Map-B comparison families: noise, faithfulness/conflict, conflict-vs-poisoning, security, and calibration/control. | Case-study worksheet; 60-pair the 60-pair validation validation pool. | Makes the five re-analyses a defined audit sample rather than isolated commentary. |
| Coding | Code each paper by failure, assumption, intervention, metric, and action. | Five-coordinate codebook; Map-B fields; pair-level worksheets. | Separates shared topics from comparable claims. |
| Verdict rule | ‘valid‘ only if the metric answers the same question under aligned evidence state and action; ‘partial‘ for complementary mechanisms; ‘invalid‘ for different games; ‘under-specified‘ for missing coordinates. | Pairwise re-analysis worksheet; the 60-pair validation gold labels. | Gives reproducible judgment criteria. |
| Disagreement handling | Use independent expert/gold verdicts for 60 unseen pairs; inspect coordinate-level errors. | Validation: 360 judgments, 180 gold votes, panel consensus. | Tests whether the tuple changes expert judgments under a blinded task. |
| Output | Report changed recommendation rather than new system leaderboard. | Main Table 3. | This is a comparability meta-evaluation, not a re-run of original RAG systems. |

Table 27: Re-analysis protocol. The five main examples follow a fixed sampling, coding, verdict, and disagreement-handling procedure; broader survey detail is left to the appendix.

| Audit quantity | Observed distribution | Agreement / change | Interpretation |
|---|---|---|---|
| Sampling frame | 60 unseen paper pairs: 32 within-family pairs and 28 cross-family stress pairs. | 13 strata spanning adversarial, calibration, distraction, evaluation, conflict, modality, noise, survey/background, and cross-family contrasts. | Extends the five illustrative re-analyses into a stratified pair-comparison audit. |
| Panel-gold verdicts | 36 partial, 24 invalid, 0 valid, 0 under-specified. | Gold Fleiss $\kappa = 0.772$; 50/60 unanimous pairs. | The audit tests hard partial/invalid triage, not all possible verdict categories. |
| Condition-level accuracy | Control 0.567; protocol 0.883. | Protocol–control = +0.317, clustered bootstrap 95% CI [0.192, 0.439]. | Tuple-guided annotators are closer to panel gold on this hard-pair sample. |
| Pair-level recommendation changes | Protocol corrects control on 19 pairs; control corrects protocol on 1; both correct on 34; both wrong same on 6. | 53/60 pair-level majority verdicts are correct under protocol vs. 35/60 under control. | The tuple changes many survey-style pair judgments from over-broad invalid/partial choices to panel-consistent verdicts. |
| Coordinate error profile | Protocol wrong verdicts 21/180; control wrong verdicts 78/180. | Protocol misses fewer failure, assumption, intervention, and action coordinates; it sometimes over-tags assumptions. | The protocol improves traceability and reduces coordinate omissions, not just verdict agreement. |

Table 28: Systematic 60-pair comparison audit. The sample is stratified over Map-B comparison families and adjacent-survey-style grouping risks. It supports claims about hard partial/invalid pair triage, not field-level rates over all robust-RAG papers.

| **Reliability-check metric** | **Value** |
|---|---|
| Paper pairs | 75 |
| Likely comparable / near-miss / negative | 25 / 25 / 25 |
| Agreement | 97.3% |
| Cohen's $\kappa$ | 0.94 |
| Comparable by source 1 | 25 |
| Comparable by source 2 | 27 |

Table 29: Decision-procedure reliability check. Pairwise comparability labels were derived from two human coding sources by requiring aligned coded fields; this supports reproducibility of coded verdicts, not independent expert-use validation.

| Apparent conclusion | What the framework changes | Non-trivial finding | Implication |
|---|---|---|---|
| "Method A is more robust to noise than method B." | Aligns noise type, rank position, gold-evidence availability, and intervention point. | Some methods solve generator distraction while others solve retriever defects; their scores are not competing estimates of the same property. | Report noise source and intervention point before ranking methods. |
| "Defense X lowers attack success." | Separates prevention, demotion, detection, traceback, and recovery. | A defense can improve traceback without lowering user-facing attack success, or certify bounded corruption without handling prompt-bearing documents. | Security papers should report outcome type, not only attack success rate. |
| "Calibration improves reliability." | Separates confidence in answer correctness from evidence sufficiency and action utility. | A calibrated answerer can still answer from insufficient evidence; an abstaining system can lower risk while reducing coverage. | Calibration metrics should be paired with evidence and action metrics. |
| "Benchmark Y is harder." | Compares evidence state, required action, cost budget, and metric target. | A benchmark can be hard because retrieval fails, because evidence conflicts, or because the correct action is refusal; these failures should not be pooled. | Leaderboards should expose diagnostic sub-scores instead of one robust-RAG score. |

Table 30: Distilled classes of non-trivial findings produced by the comparability framework. Main Table 25 gives the concrete paper-pair cases; this appendix table groups them into reusable interpretation rules.

| Main claim | Human-checked evidence basis | What the evidence supports | Boundary |
|---|---|---|---|
| Irrelevant or noisy context can harm generation. | Irrelevant-context robustness, RAAT, RGB, and counterfactual-noise papers were checked at method/evaluation level. | Noise affects both retrieval quality and generator context use. | Does not imply one robustness score covers all noise types. |
| Knowledge conflict differs from adversarial corruption. | Conflict papers and poisoning papers were checked for source of contradiction and evaluation target. | Natural disagreement and attacker-created contradiction require different assumptions. | Conflict-resolution results should not be read as poisoning-defense results. |
| RAG attacks require capability normalization. | Poisoning, backdoor, prompt-injection, traceback, and certification papers were checked for attacker access and metric target. | Attack/defense results are incomparable without corpus write access, query control, retriever visibility, and budget. | The paper does not rank attacks by real-world prevalence. |
| Calibration, abstention, and retrieval control are distinct. | Sufficient-context, uncertainty, adaptive-retrieval, and calibration papers were checked for decision target. | Confidence estimation, evidence sufficiency, retrieval triggering, and abstention optimize different quantities. | Does not validate a single universal control policy. |
| Action-trace evaluation is a benchmark agenda. | Existing benchmarks were checked for answer, evidence, action, cost, and threat fields. | Existing benchmarks cover pieces of the proposed schema. | No new leaderboard or empirical system ranking is claimed. |

Table 31: Core-claim verification ledger. LLM-assisted citation audits were used only as internal triage; the claims above are stated at the level supported by human inspection of central cited papers.

| Overlapping work | Typical interpretation in adjacent surveys | Reinterpretation under evidence reliability + action control | Different conclusion enabled |
|---|---|---|---|
| RGB / RAG evaluation benchmarks | A benchmark measuring multiple RAG abilities. | A diagnostic split between noisy evidence, negative rejection, integration, and counterfactual evidence states. | A single RAG score hides which system action failed: ignore, integrate, reject, or expose conflict. |
| Irrelevant-context robustness | A robustness method or context-filtering problem. | A generator-selectivity failure conditioned on retrieved but harmful context. | It should not be compared directly with retriever training unless the intervention point and context hardness are fixed. |
| Sufficient Context / answerability work | An abstention or uncertainty benchmark. | A control-policy problem: evidence sufficiency determines answer vs re-retrieve vs abstain. | Abstention accuracy is incomplete without evidence sufficiency and action-cost labels. |
| PoisonedRAG / RAG poisoning | A security attack surface. | An integrity failure whose strength depends on corpus write access, retrieval success, generator access, and insertion budget. | Attack success rates are not comparable until capability fields are normalized. |
| RAGTruth / hallucination evaluation | A faithfulness or hallucination annotation resource. | A separation between answer correctness, claim support, citation support, and evidence sufficiency. | Correct answers can still fail evidence reliability; citation faithfulness and answer accuracy should be reported separately. |
| Self-RAG / adaptive retrieval | A retrieval-augmented generation method. | A system-control policy that chooses retrieve, critique, answer, or continue under uncertainty. | The relevant evaluation target is not only answer quality but whether the policy chose the right action at the right cost. |

Table 32: How the same papers are interpreted differently under the proposed framework. This table grounds the novelty claim in concrete reinterpretations rather than broad positioning statements.

| Dimension | Distribution in the 342-paper adjudicated Map B |
|---|---|
| Publication year | 2025: 170; 2024: 92; 2026: 65; 2023: 12; 2021: 2; 2022: 1 |
| Source / venue type | arXiv/preprint: 207; venue/proceedings-linked: 126; workshop/challenge: 5; other/unclear: 4 |
| Primary failure category | adversarial corruption: 120; knowledge conflict: 57; calibration/control: 55; evaluation/benchmark: 43; noisy retrieval: 34; distraction: 18; extension layer: 7; survey/background: 6; OOD/uncertainty: 2 |
| Method family | defense/mitigation: 119; attack: 75; benchmark/evaluation: 67; control/calibration: 27; method/evaluation: 21; analysis: 21; survey background: 6; conflict modeling: 4; method: 2 |
| Intervention point | generator: 81; corpus/KB: 67; evaluation: 51; context assembly: 48; mixed: 43; system control: 36; retriever/context: 15; not applicable: 1 |

Table 33: Descriptive corpus statistics for the canonical 342-row adjudicated Map B. These are map statistics, not field-wide prevalence estimates. Map A remains the IAA/adjudication snapshot.

| Evidence stratum | Count |
|---|---|
| arXiv / preprint | 207 |
| venue / proceedings-linked | 126 |
| workshop / challenge | 5 |
| other / unclear | 4 |

Table 34: Source strata in the canonical 342-row adjudicated Map B. Venue-linked coverage is now substantial relative to Map A, but the map remains a scoping corpus rather than a venue-complete census.

| | Noise | Distract | Conflict | Attack | OOD | Control | Eval. | Ext. |
|---|---|---|---|---|---|---|---|---|
| Noise | 86 | 39 | 27 | 10 | 3 | 26 | 19 | 0 |
| Distract | 39 | 51 | 11 | 3 | 0 | 5 | 11 | 0 |
| Conflict | 27 | 11 | 104 | 22 | 5 | 20 | 34 | 1 |
| Attack | 10 | 3 | 22 | 135 | 1 | 10 | 21 | 0 |
| OOD | 3 | 0 | 5 | 1 | 40 | 34 | 10 | 0 |
| Control | 26 | 5 | 20 | 10 | 34 | 85 | 22 | 2 |
| Eval. | 19 | 11 | 34 | 21 | 10 | 22 | 90 | 1 |
| Ext. | 0 | 0 | 1 | 14 | 0 | 2 | 1 | 26 |

Table 35: Category overlap in the 342-row adjudicated Map B. Extension-layer labels merge the historical `modality_scenario` and `multimodal_agentic_code` names; counts are map diagnostics, not field prevalence.

| **Method family** | Noise | Distract | Conflict | Attack | Control | Eval. | Ext. | Survey |
|---|---|---|---|---|---|---|---|---|
| defense/mitigation | 27 | 12 | 33 | 37 | 9 | 0 | 1 | 0 |
| attack | 0 | 0 | 0 | 75 | 0 | 0 | 0 | 0 |
| benchmark/evaluation | 3 | 1 | 11 | 8 | 4 | 40 | 0 | 0 |
| control/calibration | 0 | 0 | 0 | 0 | 26 | 0 | 0 | 0 |
| method/evaluation | 3 | 1 | 1 | 0 | 8 | 2 | 6 | 0 |
| analysis | 1 | 4 | 8 | 0 | 6 | 1 | 0 | 0 |
| other/background | 0 | 0 | 4 | 0 | 2 | 0 | 0 | 6 |

Table 36: Primary method-family by primary failure-type cross-tabulation over the 342-row adjudicated Map B. The table describes the map, not field-wide prevalence.

# E Detailed Audit Artifacts

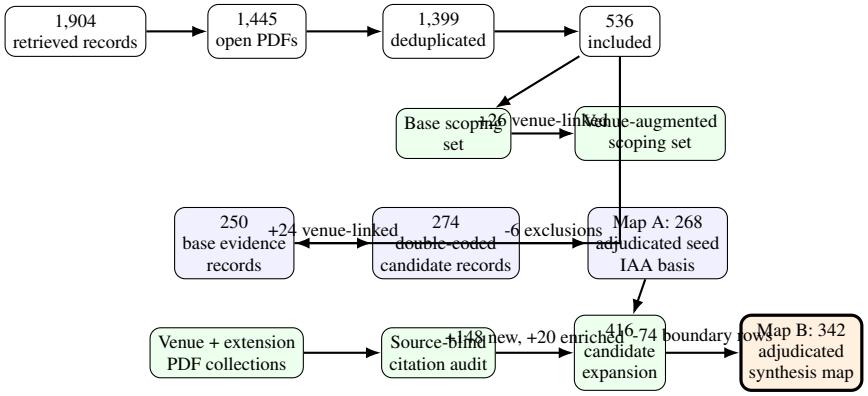



Figure 6: Corpus lineage. The 416-row venue-expanded candidate map is second-coded and adjudicated into Map B (342 rows), the canonical synthesis evidence map. Map A (268 rows) is the double-coded adjudicated seed used for IAA and adjudication audit. The base and venue-augmented scoping sets are used for boundary checks, while a separate 300-paper source-blind citation audit over the venue and extension PDF collections decides which papers enter the main text, case studies, or appendix.

| Name | Count | Construction rule | Role in this paper |
|---|---|---|---|
| Retrieved/open corpus | 1,904 → 1,445 → 1,399 → 536 | Search records, open PDFs, deduplicated candidates, then relevant included papers. | Candidate universe; not a prevalence denominator. |
| Base scoping set | 300 papers | Core reading set plus representative supporting work across failure categories. | Early scoping scaffold; distinct from the source-blind citation audit. |
| Venue-augmented scoping set | 326 papers | Base scoping set plus 26 venue-linked additions from follow-up triage. | Boundary and recall-risk check; not the evidence-map denominator. |
| Map-A candidates | 274 records | 250 base evidence records plus 24 venue-linked evidence additions. | Double-coded by two human sources before adjudication. |
| Map A adjudicated seed | 268 records | Map-A candidates after six final exclusions. | IAA/adjudication basis and provenance snapshot, not the canonical synthesis map. |
| Venue PDF collection (200 papers) | 200 local PDFs | Venue/proceedings-linked open PDFs from ACL-family, ICLR, ICML, NeurIPS, and related sources. | Venue-backed reading/audit collection; future census substrate. |
| Extension PDF collection (100 papers) | 100 local PDFs | Multimodal, graph, table, agentic/tool, code/symbolic, domain, and long-context RAG PDFs. | Extension-layer coverage, used only when tied to evidence-state/action-control claims. |
| Source-blind citation audit | 300 decisions | Source-blind human citation decisions over the venue and extension pools: 115 main, 25 case-study, 110 appendix, 22 background, 28 exclude. | Human audit for which venue/extension papers enter main text, case studies, appendix, or exclusion. |
| **Map B adjudicated synthesis map** | **342 records** | Map A (268) plus 74 retained venue/extension additions after second coding and final adjudication; 74 candidate additions were excluded. | **Primary evidence map for the current synthesis, source-strata analysis, and venue sensitivity.** |

Table 37: Canonical corpus story. All synthesis counts in the main text refer to the 342-row adjudicated Map B unless a table explicitly says it reports Map A IAA/adjudication evidence or the 416-row pre-adjudication candidate map.

| Audit item | Observed count | Rule / source | Use in main claims |
|---|---|---|---|
| Screening flow | 1,904 records → 1,445 open PDFs → 1,399 deduplicated candidates → 536 included | Keyword and curated-list retrieval over arXiv, ACL Anthology, OpenReview, USENIX, AAAI/ACM pages with open versions; deduplicated by normalized title and identifiers. | Defines the initial corpus snapshot; not used as field-wide prevalence. |
| Map A adjudicated seed | 268 records | Human-coded candidate set after round-2 adjudication. Source strata: 42 venue/proceedings-linked, 222 arXiv-only/emerging preprints, and 4 other records. | Preserved as the original adjudicated snapshot and IAA basis; not the canonical synthesis denominator. |
| Map B adjudicated synthesis map | 342 records | Map A plus 74 adjudicated venue/extension additions retained from the 148 candidate additions; 74 candidate additions were excluded after second coding and final adjudication. Source strata: 126 venue/proceedings-linked, 207 arXiv/preprint, 5 workshop/challenge, 4 other/unclear. | Current survey evidence map for source-stratified synthesis; venue-linked papers form a substantial backbone but not a venue-complete census. |
| Venue recall audit | 51 venue-linked candidates surfaced; 26 added to scoping set; 24 added to evidence extension; 9 excluded; 2 duplicates | ACL Anthology scan plus DBLP/OpenAlex/OpenReview/PMLR-style metadata passes and manual triage. | Quantifies recall risk rather than claiming venue-complete coverage. |
| Venue-linked backbone by primary label | 126 venue/proceedings-linked rows: 28 evaluation, 26 adversarial corruption, 22 noisy retrieval, 20 calibration/control, 16 knowledge conflict, 6 distraction, 6 extension-layer, and 2 OOD/uncertainty rows. | Counted from adjudicated Map B after second coding and final adjudication. | Shows that the text/document perspectives have venue-linked support while extension rows remain typed-evidence stress tests; method rankings remain claim- and assumption-specific. |

Table 38: Main-text corpus and source-strata audit. Detailed query logs, exclusions, and reconciliation tables remain in the appendix and supplement.

| Collection item | Count / location | Construction | Use and limitation |
|---|---|---|---|
| Venue-linked PDF collection | 200 local PDFs | Open PDFs copied or downloaded from ACL/TACL, ICLR, ICML, NeurIPS, and related proceedings-linked sources. | Strengthens the venue-backed reading set and feeds the source-blind citation audit; it is a PDF collection, not itself the Map-B denominator. |
| Selection sources | 97 foundations; 103 accepted/proceedings papers | Deduplicated by normalized title and filtered by RAG, retrieval, faithfulness, security, calibration, and evaluation terms. | Reduces dependence on arXiv-only discovery, but still reflects open-PDF and keyword bias. |
| Venue concentration | 49 NeurIPS; 38 ICLR; 16 ICML; plus ACL-family venues | Manifest records venue, year, source URL, local path, bytes, and SHA-256 checksum for each PDF. | Provides an auditable venue-linked supplement; manual proceedings-level inclusion/exclusion coding remains future work. |
| Machine-readable files | Manifest, summary, builder script | Built by the venue-200 collection script. | The PDF directory is local due to size; manifests and script are included in the release package. |

Table 39: Venue-linked 200-paper PDF collection. This collection addresses venue-backed coverage weakness by creating a local open-PDF corpus for citation audit and future venue-complete follow-up, while preserving the distinction between downloaded PDFs, the 416-row candidate expansion, and the canonical 342-row adjudicated Map B.

| Map-B field group | Coverage | Coding / audit source | Use and caveat |
|---|---|---|---|
| Canonical identifiers and source flags | 342/342 title, year, URL, source stratum, Map-A membership, Map-B version, core-reading flag | Deterministic merge from Map A, the venue/extension pools manifests, source-blind citation-audit decisions, and second-coding adjudication. | Defines the canonical adjudicated row set and source strata. |
| Core taxonomy fields | 342/342 failure type, taxonomy categories, method family, intervention point, contribution note, limitation note | Map-A adjudicated labels for original rows; second coding adjudicated labels for retained additions. | Used for Map-B synthesis and sensitivity counts. |
| Task / metric metadata | High coverage after adjudication | Map-A labels plus human-audit rationales and manifest metadata for retained additions. | Missing rows are not imputed for metric-level claims. |
| Author / venue metadata | Source/venue strata complete; author metadata incomplete for some additions | Proceedings metadata and paper manifests. | Venue/source strata are complete; author normalization remains an artifact-release cleanup item. |
| Human-audited rows retained in Map B | 94 retained rows with human citation-audit metadata | 74 retained additions plus 20 enriched Map-A rows. | Citation-audit metadata supports source-stratified synthesis; 74 excluded candidate additions remain in the release as boundary rows. |
| Second-coding/adjudication boundary | Map A: 274 candidates $\rightarrow$ 268 adjudicated rows; Map-B additions: 148 candidates $\rightarrow$ 74 retained additions | Map A has two-source independent coding and final adjudication; Map-B additions have independent second coding and final adjudication. | Second-coding agreement is strong, and adjudication is now applied to the canonical 342-row Map B. |

Table 40: Coding protocol and field coverage for the 342-row adjudicated Map B. The table separates Map-A adjudicated IAA, source-blind citation-audit metadata, and the second-coding/adjudication pass over the 148 candidate additions.

| Field / decision | Map-A evidence | Map-B addition evidence | Adjudication effect | Claim use |
|---|---|---|---|---|
| Primary failure type | 274 candidates double-coded (268 retained as Map A after adjudication); $\kappa = 0.96$. | second-coding $\kappa = 0.719$; final-label agreement $\kappa = 0.743/0.874$. | Candidate labels changed for many additions; final labels are adjudicated. | Supports comparison-family grouping, not field prevalence. |
| Method family | Map-A $\kappa = 0.86$. | second-coding $\kappa = 0.762$; final-label agreement $\kappa = 0.818/0.915$. | Broad candidate method labels were often corrected. | Used for re-analysis and method-role distinctions. |
| Intervention point | Map-A $\kappa = 0.92$. | second-coding $\kappa = 0.724$; final-label agreement $\kappa = 0.807/0.863$. | Pipeline-location labels required adjudication. | Used for comparability verdicts. |
| Inclusion / membership | Map-A include decision $\kappa = 0.85$. | second-coding inclusion $\kappa = 0.769$; final-label agreement $\kappa = 0.905/0.865$. | 148 candidate additions $\rightarrow$ 74 retained, 74 excluded. | Defines canonical 342-row Map B; excluded rows are release-only boundary evidence. |
| Candidate-label instability | – | 139/148 additions required adjudication; 125/148 changed on at least one field vs. candidate labels. | High override rate is reported rather than hidden. | Prevents treating candidate expansion statistics as final synthesis statistics. |

Table 41: Main-text reliability summary for Map B. Agreement numbers for the second coding are reported against both each other and the final adjudicated labels; the table makes clear that the 416-row candidate expansion is not the canonical map.

| Adjudication category | Count | Typical reason | Consequence |
|---|---|---|---|
| Accept coder consensus | 114/139 disagreement rows | the second coders agreed that many candidate additions were general RAG, efficiency, domain adaptation, or application papers rather than robust-RAG evidence. | Candidate expansion statistics are not used directly; boundary rows are released but excluded from canonical Map B. |
| Keep candidate label | 14/139 | Candidate label remained appropriate despite one coder disagreement, usually when the paper clearly addressed a robust-RAG failure or benchmark axis. | Shows adjudication did not mechanically follow coders. |
| Resolve mixed labels | 7/139 | Coders and candidate labels captured different valid aspects, e.g., control plus evaluation or extension plus retrieval. | Final label chooses the primary comparison role; secondary evidence remains in notes. |
| Accept coder 2 | 4/139 | the second coder better matched the paper's primary contribution after full adjudication. | Small number of one-coder overrides are recorded explicitly. |
| Retained additions | 74/148 | Robustness, evaluation, control, conflict, or security contribution survives adjudication. | Included in canonical 342-row Map B. |
| Excluded additions | 74/148 | Boundary/application/general-RAG papers, often venue-linked, did not directly study unreliable evidence. | Excluded from synthesis counts but retained as release artifacts. |

Table 42: Map-B candidate-addition override audit. The high adjudication rate is treated as evidence that the 416-row candidate expansion was too inclusive; the canonical map uses the 342-row adjudicated version.

| Stage | Count | Rule |
|---|---|---|
| Initial retrieval | 1,904 records | Keyword search over arXiv, ACL Anthology, OpenReview, USENIX, AAAI/ACM landing pages where open versions were available, plus curated robust/trustworthy RAG paper lists. |
| Open full-text corpus | 1,445 PDFs | Downloaded only open PDFs; removed failed downloads, inaccessible files, and non-PDF artifacts. |
| Deduplicated candidates | 1,399 papers | Deduplicated by normalized title and arXiv/proceedings identifiers; retained the most complete metadata record when preprint and proceedings versions overlapped. |
| Relevance screening | 536 included papers | Included papers that study RAG robustness, reliability, uncertainty, calibration, faithfulness, attack/defense, or evaluation under unreliable evidence; excluded application-only RAG papers. |
| Core pool | 112 papers | Papers directly shaping the taxonomy, threat/failure model, intervention mechanisms, or evaluation framework. |
| Core reading set | 100 papers | Core papers used as main narrative anchors; the remaining 12 core-pool papers stay in the broader scoping set. |
| Base scoping set | 300 papers | Core papers plus representative supporting work across all taxonomy branches before venue follow-up triage. |
| Venue-augmented scoping set | 326 papers | Base scoping set plus 26 venue-linked additions after follow-up triage. |
| Candidate evidence records | 274 papers | Base evidence map of 250 papers plus 24 venue-linked additions, double-coded and adjudicated. |
| Map A adjudicated seed | 268 papers | Candidate evidence records retained after final adjudication; used for IAA/adjudication provenance. |
| Map-B candidate expansion | 416 papers | Map A plus 148 source-blind human-citation-audit additions and 20 enriched original rows; used for second-coding review. |
| Map B adjudicated synthesis map | 342 papers | Map A plus 74 retained additions after second coding and final adjudication; used for current synthesis statistics. |

Table 43: Screening-flow summary for the current corpus snapshot. The supplementary material provides exact query strings, source-wise counts, deduplication rules, taxonomy codebook, per-paper metadata, exclusion log, Map-A adjudication records, and the adjudicated Map-B evidence map.

| Set | N | Audit artifact | Selection rule and role |
|---|---|---|---|
| Retrieved records | 1,904 | `search_protocol.json` | Initial keyword and curated-list retrieval before PDF availability, deduplication, or relevance screening. |
| Open full-text PDFs | 1,445 | `pdfs/` | Records with open PDFs available for full-text inspection; inaccessible or non-PDF artifacts removed. |
| Deduplicated candidates | 1,399 | `all_reviewed_deduped.csv` | Normalized-title and identifier deduplication; one record retained for preprint/proceedings overlaps. |
| Relevant included papers | 536 | `all_reviewed_deduped.csv` | Papers with decision `core` (112) or `include` (424); background-adjacent papers (289) and excluded papers (574) are tracked separately. |
| Core pool | 112 | `decision=core` in `all_reviewed_deduped.csv` | Papers judged central to taxonomy, threat/failure models, intervention mechanisms, or evaluation design. |
| Core reading set | 100 | `core100_full.csv` | Main narrative anchors. The 12 core-pool papers not in this set remain in the base scoping set rather than being discarded. |
| Base scoping set | 300 | `survey300_full.csv` | Core pool plus representative supporting work across failure categories, method families, and evaluation settings before venue follow-up triage. |
| Venue-augmented scoping set | 326 | `survey_set_venue_audited.csv` | Base scoping set plus 26 venue-linked papers added after follow-up triage. |
| Base evidence map | 250 | `evidence_map.csv` | Original structured map used before venue follow-up triage. |
| Candidate evidence records | 274 | round2 full adjudication CSV | Base map plus venue-linked additions, double-coded by humans and adjudicated. |
| Map A adjudicated seed | 268 | `evidence_map_adjudicated.csv` | Candidate evidence records with final `include_in_evidence_map=1`; used for IAA/adjudication provenance. |
| Source-blind citation audit | 300 | `citation_annotations.csv` | Source-blind human citation decisions over the venue and extension pools; distinct from the base scoping set. |
| Map-B candidate expansion | 416 | candidate expansion CSV | Map A plus 148 candidate additions supported by the source-blind citation audit and 20 enriched original rows; used for second coding. |
| Map B adjudicated synthesis map | 342 | adjudicated Map-B CSV | Map A plus 74 retained additions after second-coding adjudication; used for current synthesis statistics. |

Table 44: Corpus-set ledger. Exact memberships are released as CSV files for Map A, the 416-row candidate expansion, and the 342-row adjudicated Map B.

| Artifact / claim source | Verification performed | What is coding- or metadata-assisted | How it is used in the paper |
|---|---|---|---|
| Main narrative claims and central citations | Author inspection of full paper text for papers used to support central claims, tables, and figures. | Initial prioritization and category suggestions were LLM-assisted. | Used for qualitative synthesis and case-study comparisons. |
| Core reading set | 100 papers curated from the 112-paper core pool after full-text relevance checks for narrative fit. | Core-pool creation used relevance scores and pre-labels as triage signals. | Used as the main reading backbone; not claimed to be a reproducible top-100 ranking without author judgment. |
| Map A, 268-paper adjudicated seed | Metadata and structured fields manually cleaned; venue-candidate triage completed; 274 candidate records independently double-coded and adjudicated into 268 included evidence-map papers. | Initial candidate discovery and pre-labeling were LLM-assisted, but final Map-A labels use human adjudication for coder disagreements. | Used for IAA/adjudication provenance, with the caveat that it is not the current synthesis denominator. |
| Map-B candidate expansion, 416 rows | Map A merged with source-blind human citation audit over the venue/extension pools; 148 new candidate rows added and 20 original rows enriched. | Candidate additions use human citation-audit decisions and deterministic field completion before second coding. | Used as the candidate pool for second coding, not as the final synthesis denominator. |
| Map B, 342-paper adjudicated synthesis map | Two coders independently second-coded all 148 additions; final adjudication retained 74 and excluded 74. | The retained additions use adjudicated labels; excluded boundary rows remain available in the release package. | Used for current source-stratified synthesis and venue sensitivity; still not a field census. |
| Human double-coding package | Two anonymized human sources independently coded all 274 candidate evidence records before discussion; round-2 agreement is strong across taxonomy, intervention, evidence-strength, relevance, core, and inclusion fields. | Disagreements were resolved in the final adjudication file. | Used to report pre-adjudication IAA and produce final adjudicated evidence-map labels. |
| Venue metadata pass and triage | ACL Anthology title/abstract scan plus DBLP/OpenAlex query logs, title matching, and follow-up triage decisions. | Discovery is metadata-based; inclusion/exclusion decisions are author triage rather than independent double coding. | Used to update venue-linked coverage and quantify remaining systematic-review limits. |

Table 45: Verification and annotation provenance. The table separates full-text author inspection, coding-assisted labels, metadata-derived venue candidates, and non-adjudicated consistency diagnostics.

| Item | What was done | What was hidden from annotators | What we do not claim |
|---|---|---|---|
| Round-2 coders | Two anonymized human annotation sources independently coded the same 274 candidate evidence records. | Machine pre-labels, prior evidence-map labels, previous adjudication logs, and the other source's labels. | We do not reveal personal identities in the submission; anonymized provenance is reported instead. |
| Training / codebook | Coders used the human double-coding protocol and codebook defining scope, failure type, method family, intervention point, evidence strength, and inclusion fields. | Prior LLM rationales and model confidence scores. | We do not claim the codebook is final for all RAG subareas; adjudication notes can still motivate revisions. |
| Independence | Round-2 IAA was computed before adjudication using independently completed CSV files. | Post-discussion labels were not used for IAA. | We do not use adjudicated labels to inflate agreement. |
| Adjudication | A final adjudication file resolved 469 field-level disagreements and produced the 268-paper Map-A seed. | Source labels were preserved and not overwritten. | We do not apply Map-A IAA to the 148 later Map-B additions. |
| Source-blind citation audit | One source-blind human citation-audit batch coded 300 papers from the venue/extension pools; 168 audited rows entered or enriched Map B after de-duplication and relevance checks. | Paper IDs were source-normalized and the audit did not use Map-A disagreement labels. | We do not claim 416-row double-coding or inter-annotator agreement for the venue-expanded additions. |

Table 46: Annotation transparency for the human coding round. The table reports what can be audited from the blind annotation package without disclosing annotator identities.

| Claim | Status under venue-linked / peer-reviewed evidence | What changes without preprints | Interpretation |
|---|---|---|---|
| Retrieved evidence can harm generation even when relevant evidence exists. | Holds: supported by venue-linked work on irrelevant/noisy context, noise-robust training, counterfactual noise, and imperfect retrieval. | The range of stress-test variants shrinks. | This is a stable mechanism-level claim. |
| RAG security results require attacker-capability normalization. | Holds partially: venue-linked security work covers poisoning, traceback, membership inference, and backdoor-style corpus attacks. | Many fine-grained attack variants and defenses remain preprint-heavy. | The capability matrix is stable; detailed landscape counts are emerging trends. |
| Calibration, abstention, and retrieval control are distinct decisions. | Holds conceptually with venue-linked answerability, relevance-assessment, and calibration/control papers. | Specific method rankings and benchmark prevalence become sparse. | The conceptual separation remains; empirical prevalence is not settled. |
| The evidence map is attack- and benchmark-heavy. | Does not become a field-wide claim. | Under peer-reviewed-only evidence, the sample is much smaller and more evaluation-heavy. | Treat distributional cross-tabs as corpus diagnostics, not prevalence estimates. |
| Action-trace benchmark blueprint should score answer, evidence state, action, and cost. | Supported as a design synthesis, not validated as a benchmark standard. | The proposal becomes less empirical and more normative. | Keep as a research agenda unless a full benchmark is built and evaluated. |

Table 47: Peer-reviewed / venue-linked sensitivity analysis. The table states which conclusions survive when preprint-only evidence is downweighted and which remain hypothesis-generating.

| Conclusion tested after downweighting preprints | Venue-linked support in map | What remains supported | What remains trend evidence |
|---|---|---|---|
| Noise/distraction are not reducible to retriever recall. | 22 primary noisy-retrieval and 6 distraction venue-linked rows, plus context-compression/reranking benchmarks. | Context construction, ranking, compression, and generator selectivity all affect robustness. | Exact method rankings across retrievers, generators, and context budgets. |
| Natural conflict and poisoning are different comparison games. | 16 primary knowledge-conflict and 26 adversarial-corruption venue-linked rows. | Source reconciliation and attacker-created integrity failures require different assumptions and metrics. | Transfer from conflict-resolution methods to adaptive poisoning defenses. |
| RAG security results require capability normalization. | 26 primary adversarial-corruption venue-linked rows, including poisoning, privacy/extraction, and defenses. | Attack success is uninterpretable without corpus access, query control, retriever visibility, insertion budget, objective, and defense action. | Detailed cross-threat leaderboards across heterogeneous attacker capabilities. |
| Calibration, abstention, and retrieval control are separable. | 20 primary calibration/control venue-linked rows plus answerability and adaptive retrieval work. | Confidence target, evidence sufficiency, coverage, retrieval cost, and action utility should be reported separately. | Frontier-model control policies and multi-action cost tradeoffs across domains. |
| Answer accuracy alone is insufficient for robust-RAG evaluation. | 28 primary evaluation/benchmark venue-linked rows. | Evidence state, support, threat model, action policy, and cost are needed to interpret results. | A full action-trace benchmark remains future work. |
| Extension-layer RAG requires typed evidence states. | 6 primary extension-layer venue-linked rows. | Visual, graph, table, memory, tool, code, and domain evidence require modality-specific grounding and action labels. | Unified evaluation across text, multimodal, graph, table, and agentic RAG remains immature and is not treated as co-equal coverage. |

Table 48: Peer-reviewed / venue-linked sensitivity in the main text. Counts refer to primary labels in the 342-record adjudicated venue-expanded evidence map.

| Robust-RAG setting | Venue-backed basis | Recommended comparison target | Action policy to report | Boundary condition |
|---|---|---|---|---|
| Benign noisy retrieval / distraction | 22 noisy-retrieval and 6 distraction venue-linked rows; IrrelCtx, RAAT, RECOMP, RankRAG, RGB. | Compare filters, rerankers, compressors, training methods, and readers only under matched negative type, rank, gold evidence, and generator. | answer, filter, rerank/compress, abstain/retrieve-more. | More context can help when gold evidence is missing; ranking claims require a shared harness. |
| Faithfulness / attribution / conflict | 16 knowledge-conflict venue-linked rows plus RAGTruth, FaithEval, Trust-Align, reconciliation work. | Compare span support, claim support, source conflict, and final answer correctness as different targets. | answer, verify, expose conflict, refuse/defer. | Citation support is not sufficient evidence of conflict robustness. |
| Adversarial RAG security | 26 adversarial-corruption venue-linked rows including poisoning, privacy/extraction, and defenses. | Compare attacks/defenses inside one capability envelope: corpus access, query control, budget, retriever visibility, objective. | prevent, demote, detect, quarantine, trace, recover, or limit leakage. | Cross-threat leaderboards are invalid unless threat model and clean utility align. |
| Calibration / retrieval control / abstention | 20 calibration/control venue-linked rows plus answerability and adaptive retrieval work. | Compare confidence targets, sufficiency labels, retrieval triggers, and action utility separately. | answer, retrieve again, clarify, abstain, verify, escalate. | Calibration error alone does not define a policy. |
| Evaluation resources | 28 evaluation/benchmark venue-linked rows including RGB, RAGTruth, RAGAS, ARES, BEIR, BERGEN, RAGLAB. | Use diagnostic scorecards rather than one robust-RAG score. | required action label and cost/coverage reporting. | Toolkits, annotated corpora, retriever benchmarks, and system harnesses play different roles. |

Table 49: Venue-stratified positive recommendations. These recommendations are anchored in the venue/proceedings-linked subset of the adjudicated Map B and state the threat model, metric target, and action policy under which comparisons are meaningful.

| Pass | Scope | Observed yield | Interpretation |
|---|---|---|---|
| ACL Anthology metadata scan | ACL, EMNLP, NAACL, COLING; 2020–2026; full anthology BibTeX with abstracts | 12,924 entries scanned; 181 RAG candidates; 30 robust-RAG candidates | A reproducible proceedings-level metadata scan for core NLP venues, not just backfilled metadata. |
| DBLP/OpenAlex metadata queries | 15 robust-RAG query strings over venue-linked metadata for SIGIR, WWW, WSDM, KDD, USENIX Security, CCS, NDSS, ICLR, NeurIPS and overlaps | 31 query rows; 51 unique venue-linked candidate titles after deduplication | Surfaced missing venue-linked candidates; all 51 were subsequently triaged. |
| Follow-up venue triage | 51 venue-linked candidates | 23 new evidence-map additions; 3 background additions; 9 exclusions; 2 duplicates; 13 already represented | Closes the unprocessed venue-candidate gap, while still not claiming complete manual proceedings coverage. |
| Map A adjudicated seed | 268 evidence-map papers | 42 venue/proceedings-linked; 222 arXiv-only or emerging preprints; 4 other strata | IAA/adjudication basis before venue expansion. |
| Map-B candidate expansion | 416 evidence-map candidates | 194 venue/proceedings-linked; 207 arXiv/preprint; 11 workshop/challenge; 4 other/unclear | Pre-adjudication candidate map after source-blind citation audit. |
| Map B adjudicated synthesis map | 342 evidence-map papers | 126 venue/proceedings-linked; 207 arXiv/preprint; 5 workshop/challenge; 4 other/unclear | Current source-stratified synthesis map after second coding and final adjudication. |

Table 50: Venue-oriented search, coverage audit, and follow-up triage summary. Full query logs, candidate titles, decisions, and limitations are in `venue_systematic_queries.csv`, `venue_systematic_candidates.csv`, `venue_coverage_audit.csv`, and `venue_triage_decisions.csv`.

| Coverage check | Count | Artifact | Interpretation |
|---|---|---|---|
| Venue-linked robust-RAG candidates found by metadata pass | 51 unique titles | candidate CSV | Candidate pool from ACL Anthology, DBLP, and OpenAlex metadata queries. |
| Already in core100 | 6 / 51 | coverage audit CSV | Core narrative captures only a small subset of venue-linked candidates. |
| Already in base scoping set | 14 / 51 | coverage audit CSV | The current scoping set misses or intentionally excludes many venue-linked candidates. |
| Already in evidence map | 13 / 51 | coverage audit CSV | Cross-tabs should be interpreted as map statistics, not field statistics. |
| Not in original base scoping set | 37 / 51 | coverage audit CSV | These papers were subsequently triaged; see Table 53. |

Table 51: Venue-coverage audit after the metadata pass. The audit quantifies residual coverage risk by matching venue-linked candidates against the original core, survey, and evidence-map sets. Files: `venue_systematic_candidates.csv` and `venue_coverage_audit.csv`.

| Recall check | Observed result | What it controls | Residual risk |
|---|---|---|---|
| ACL Anthology metadata scan | 12,924 entries scanned for ACL/EMNLP/NAACL/COLING; 181 RAG candidates; 30 robust-RAG candidates | Reduces reliance on arXiv-only discovery for core NLP venues. | Title/abstract terms can miss papers using older QA/IR terminology. |
| Cross-venue metadata pass | DBLP/OpenAlex queries yielded 51 venue-linked robust-RAG candidates | Adds IR, web, security, and ML venue-linked records. | API ranking, metadata naming, and transient query errors limit recall. |
| Follow-up triage | 51 candidates triaged; 26 added to scoping set; 24 added to evidence map; 9 excluded; 2 duplicates | Closes the known unprocessed venue-candidate gap. | Does not prove complete coverage of every proceedings volume. |
| Evidence-strength sensitivity | Main conclusions separated into venue-linked backbone vs preprint trend evidence | Prevents preprint-heavy map statistics from becoming field-level claims. | Some emerging security/control subtopics remain too new for strong consensus. |

Table 52: Venue-recall risk analysis. The venue pass improves auditability and adds missed venue-linked papers, but it is still not a manual venue-complete proceedings review.

| Triage outcome | Count | Action | Reasoning |
|---|---|---|---|
| Already represented in base evidence map | 13 | retain | Candidate was already included; metadata pass only improved venue visibility. |
| New evidence-map additions | 23 | add to evidence extension | Directly studies robust RAG, RAG evaluation, evidence reliability, security, calibration, or robust generation. |
| Scoping-set paper promoted to evidence extension | 1 | add to evidence extension | Already in the scoping set but not the base map; venue audit identified it as relevant to trust/calibration. |
| Background additions | 3 | add to venue-augmented scoping set only | Relevant boundary papers on domain, multilingual, or multimodal RAG, but not central text evidence-reliability evidence. |
| Excluded false positives / applications | 9 | exclude with reason | Application or adjacent RAG papers without robustness/evidence-reliability analysis. |
| Duplicate venue records | 2 | deduplicate | Same work already represented through another metadata record. |
| **Venue-augmented scoping set** | **326** | base 300 + 26 additions | Updated scoping set after follow-up triage. |
| **Map A adjudicated seed** | **268** | 274 candidates - 6 exclusions | IAA/adjudication snapshot after human adjudication of double-coded candidate evidence records. |
| **Map-B candidate expansion** | **416** | Map A + 148 citation-audit-supported additions | Pre-adjudication candidate map after source-blind citation audit. |
| **Map B adjudicated synthesis map** | **342** | Map A + 74 retained additions | Current evidence map for synthesis after second coding and adjudication. |

Table 53: Follow-up triage of all 51 venue-linked candidates from the metadata pass. The 37 candidates outside the original scoping set are no longer unprocessed; decisions and reasons are in `venue_triage_decisions.csv`.

| Artifact / set | Count | Count definition | Use in this paper |
|---|---|---|---|
| Map-A double-coded candidates | 274 | Candidate records coded by two human sources before final inclusion adjudication. | Used for human-human agreement and adjudication; six records were excluded from Map A. |
| Map A adjudicated seed | 268 | Included records after adjudication: 42 venue/proceedings-linked, 222 arXiv-only or emerging preprints, 2 preprints with year tags, 1 review-platform record, and 1 other/uncategorized record. | Used as the IAA/adjudication provenance snapshot, not the current synthesis denominator. |
| Venue-augmented scoping set | 326 | Original base scoping set plus 26 triaged venue-linked additions. | Used for broader scoping and boundary checks; not all rows are evidence-map records. |
| Source-blind citation audit | 300 | Human citation decisions over venue PDFs and extension PDFs. | Determines which audited papers enter the main text, case studies, appendix, background, or exclusion set; distinct from the base scoping set. |
| Map-B candidate expansion | 416 | Map A plus 148 candidate additions supported by the source-blind citation audit; 20 original rows enriched with citation-audit metadata. | Pre-adjudication expansion candidate map used for second-coding review. |
| Map B adjudicated synthesis map | 342 | Map A plus 74 retained additions after second coding and final adjudication. | Current evidence map for source-stratified synthesis and Map-B cross-tabs. |

Table 54: Reconciliation of reported corpus counts. The 342-row adjudicated Map B is the canonical synthesis map; Map A and the 274-candidate set are retained to report double-coding and adjudication provenance.

| Field | Agreement | $\kappa$ / score | Interpretation |
|---|---|---|---|
| Full-text checked | 100.0% | 1.00 | Both sources marked every evidence-map paper as full-text checked. |
| In-scope screening | 99.6% | 0.92 | Near-complete agreement; only one screening disagreement remains for adjudication. |
| Relevance level | – | 0.80 | Linear-weighted agreement is strong for exclude/background/support/core priority. |
| Primary failure type | 97.1% | 0.96 | Very strong human agreement on the main taxonomy category. |
| Method family | 89.4% | 0.86 | Strong agreement on attack, benchmark, control, method, and analysis families. |
| Intervention point | 93.4% | 0.92 | Very strong agreement on pipeline placement. |
| Evaluation type | 94.9% | 0.93 | Very strong agreement on metric/evaluation family. |
| Evidence strength | 97.8% | 0.95 | Very strong agreement on proceedings/preprint/review-platform strata. |
| Core-candidate decision | 91.6% | 0.80 | Strong agreement on whether papers should shape the core narrative. |
| Include in evidence map | 99.3% | 0.85 | Strong agreement on evidence-map inclusion. |
| Secondary labels | 79.2% exact | 0.83 Jaccard | Multilabel agreement is strong enough to support overlap analysis after adjudication. |
| Reproducibility evidence | 10.6% exact | 0.20 Jaccard | Weak free-text agreement; retained as adjudicated audit metadata, not used for primary comparison or trend claims. |

Table 55: Round-2 human-human agreement on 274 candidate evidence records before adjudication. Structured labels show strong agreement, while the free-text reproducibility field is weak and is treated only as adjudicated audit metadata. Two anonymized human annotation sources independently coded the same records; remaining disagreements were resolved to produce the 268-row Map-A adjudicated seed. These IAA values do not apply to the 148 later Map-B additions.

| Field | coder 1 vs. final | coder 2 vs. final | coder 1 vs. coder 2 | Final adjudication effect |
|---|---|---|---|---|
| Primary failure type | 0.784 / 0.743 | 0.899 / 0.874 | 0.764 / 0.719 | Final labels resolve 125 rows that changed from the candidate expansion on at least one field. |
| Method family | 0.878 / 0.818 | 0.946 / 0.915 | 0.838 / 0.762 | Broad candidate method labels were corrected toward coder-supported labels. |
| Intervention point | 0.838 / 0.807 | 0.885 / 0.863 | 0.770 / 0.724 | Pipeline-location disagreements are resolved before final Map-B synthesis. |
| Scope / inclusion | 0.953 / 0.905 | 0.932 / 0.865 | 0.885 / 0.769 | Final Map B retains 74 additions and excludes 74 boundary candidates. |

Table 56: Final-label agreement for the 148 Map-B candidate additions. Cells report agreement / Cohen's $\kappa$ against the final adjudicated label, not the pre-adjudication candidate label. This resolves the apparent instability of the candidate map: coder 1 and coder 2 agree strongly with the final labels, and excluded boundary rows are kept in the release but not in canonical Map B.

| Metric | Control | Protocol | △ (95% CI) |
|---|---|---|---|
| Experts | 15 | 15 | – |
| Pair judgments | 180 | 180 | – |
| External gold agreement | 0.800 | 0.978 | +0.178 [+0.128, +0.228] |
| Fleiss $\kappa$ | 0.589 | 0.846 | – |
| Decision time (s) | 126.1 | 121.8 | -4.3 [-13.9, +5.8] |
| Confidence (1–5) | 3.69 | 3.92 | +0.23 [+0.13, +0.32] |
| Usefulness (1–5) | 3.51 | 4.06 | +0.55 [+0.41, +0.70] |

Table 57: Blind external protocol-usability stress test. Thirty distinct annotators judged the same 12 deliberately difficult paper pairs, with 15 assigned to ordinary expert judgment and 15 assigned to the five-coordinate protocol. Agreement against independently adjudicated external gold labels is reported descriptively for this non-representative pair set; the 360 judgments are clustered within annotators and shared pairs. Confidence intervals are participant-level bootstrap intervals for protocol minus control, not population-level IID paper-pair inference.

| Pair | Control pattern | Protocol pattern | Tuple explanation | Disagreement lesson |
|---|---|---|---|---|
| P08 | 1/15 matched gold; 14/15 chose partial instead of invalid. | 13/15 matched invalid. | AstuteRAG-style benign conflict reasoning and PoisonedRAG-style attacker-created evidence differ in threat model, intervention point, metric polarity, and action. | Without the tuple, experts tended to over-group conflict and poisoning as related robustness papers. |
| P04 | 3/15 matched partial; 12/15 over-called valid. | 15/15 matched partial. | Chain-of-Note and irrelevant-context filtering share a noise family but differ in generator reasoning vs. context filtering and answer+abstain vs. filter+answer actions. | The tuple turns a tempting leaderboard comparison into a pipeline-composition claim. |
| P01 | 11/15 matched partial; 4/15 over-called valid. | 15/15 matched partial. | IrrelCtx and RAAT share answer metrics but differ in natural retrieval noise vs. constructed counterfactual noise and filter/back-off vs. answer-only actions. | The tuple exposes action-policy mismatch even when datasets and metrics overlap. |
| P09 | 13/15 matched partial; 2/15 over-called valid. | 15/15 matched partial. | PoisonedRAG and certified/robust defenses share bounded corruption but differ in attacker vs. defender role, attack success vs. certified robust accuracy, and answer/abstain/filter actions. | Even same-threat-space papers need role and action normalization before ranking. |
| P02 | 14/15 matched partial; 1/15 over-called invalid. | 15/15 matched partial. | IrrelCtx is a method paper; RGB is a benchmark artifact with negative rejection and counterfactual testbeds. | Method-vs-benchmark comparisons should ask what evaluation would make them comparable. |

Table 58: Disagreement analysis for the external expert-use stress test. The table reports illustrative real paper-pair comparisons where ordinary expert judgment and tuple-guided judgment diverged. It is usability evidence for the protocol on 12 hard pairs, not population-level validation.

| Quantity | Control | Protocol | Gold / model | Interpretation |
|---|---|---|---|---|
| Annotators / judgments | 3 / 180 | 3 / 180 | 3 gold adjudicators / 180 gold votes | Expanded validation uses unseen pairs and multiple gold adjudicators. |
| Agreement with panel gold | 0.567 | 0.883 | Panel consensus over 60 pairs | The tuple protocol improves verdict agreement on the blinded re-annotation. |
| Mean confidence | 4.07 | 3.83 | – | Protocol users are slightly less confident but substantially more aligned with panel gold. |
| Protocol minus control | +0.317, clustered bootstrap 95% CI [0.192, 0.439] | | Direction only | The agreement gain is robust in direction; we omit a point odds-ratio estimate, which is unstable with only six annotators. |
| Gold inter-agreement | – | | Fleiss $\kappa = 0.772$; 50/60 unanimous | Gold construction is stable for this hard-pair set; valid and under-specified labels are not represented. |

Table 59: Expanded 60-pair expert validation (blind re-annotation). The five-coordinate protocol improves verdict agreement with panel gold on unseen hard pairs while retaining coordinate-level error auditability. The result validates protocol-assisted triage for this hard-pair setting, not all possible comparison categories.

| Condition | Wrong verdicts | Missed gold coordinates (F/A/I/M/Act) | Extra coordinates | Use |
|---|---|---|---|---|
| Control | 78/180 | 34 / 55 / 28 / 1 / 82 | failure 12; assumption 20; metric 3; action 2 | Ordinary judgments over-call invalid in many partial cases and miss key action and assumption distinctions. |
| Protocol | 21/180 | 10 / 12 / 7 / 13 / 16 | assumption 27; failure 9; action 2 | Protocol reduces wrong verdicts and missed coordinates, though it sometimes over-tags assumptions. |

Table 60: Per-coordinate error audit for the 60-pair expert validation. Counts are over 180 judgments per condition. F/A/I/M/Act denote failure, assumption, intervention, metric, and action.

| Backbone conclusion | Venue-linked evidence base | What remains stable | What remains preprint/trend evidence |
|---|---|---|---|
| Irrelevant or noisy evidence can change generation behavior. | Irrelevant-context robustness, noise-aware training, counterfactual-noise analyses, and robust fine-tuning papers in ICLR/ACL/NAACL/SIGIR-style venues. | Robustness is not just retriever recall; generator selectivity and context construction matter. | Fine-grained noise taxonomies, long-tail stress tests, and model-specific mitigation rankings. |
| Knowledge conflict is distinct from poisoning. | Conflict-resolution and imperfect-retrieval work in TACL/COLING/EMNLP-style records, plus venue-linked conflict benchmarks. | Natural contradiction requires source reconciliation and evidence sufficiency reasoning. | Many automatic conflict-resolution methods and large-scale conflict benchmark variants. |
| RAG security comparisons require capability normalization. | USENIX/WWW/CCS/KDD-style poisoning, traceback, membership-inference, and backdoor records. | Attack success cannot be interpreted without corpus write access, query control, retriever visibility, insertion budget, and stealth. | The detailed ranking of attacks/defenses and many newly proposed bypasses. |
| Calibration and abstention are action-control decisions. | Venue-linked relevance-assessment, answerability, calibration, and RAG evaluation records. | Confidence, evidence sufficiency, retrieve/answer decisions, and abstention are separable evaluation targets. | Specific adaptive-control policies and frontier-model calibration behavior. |
| Benchmark design needs evidence and action labels. | RGB, RAGTruth, RAG evaluation libraries, long-form/context benchmarks, and challenge-style records. | Answer accuracy alone is insufficient for robust RAG evaluation. | The full action-trace blueprint remains a proposal until implemented and validated. |

Table 61: Peer-reviewed / venue-linked backbone of the analysis. These are the conclusions the paper treats as stable when preprint-only evidence is downweighted.

| Category | Definition | Inclusion rule |
|---|---|---|
| Noisy retrieval | Benign retrieval imperfection: irrelevant passages, hard negatives, missing positives, stale or low-quality evidence. | Explicitly analyzes or mitigates retrieval noise, not merely uses retrieval. |
| Distraction | Retrieved context has negative marginal utility and changes answer, confidence, or reasoning despite not supporting the answer. | Includes distractor, context-utilization, or long-context degradation analysis. |
| Knowledge conflict | Contradiction between parametric knowledge and retrieved evidence, or among retrieved sources. | Includes conflict detection, source preference, fact-level reconciliation, or contextual faithfulness. |
| Adversarial corruption | Adaptive attacker manipulates corpus, retriever, prompt template, retrieved context, memory, or tools. | Includes poisoning, prompt injection, backdoors, attack/defense benchmarks. |
| OOD / uncertainty | Query or evidence is ambiguous, underspecified, unanswerable, out-of-knowledge-base, or under distribution shift. | Includes answerability, uncertainty quantification, or selective prediction. |
| Calibration / control | System decides when to retrieve, trust evidence, answer, abstain, clarify, or escalate. | Includes adaptive retrieval, calibrated decision-making, or abstention. |
| Evaluation benchmark | Paper introduces benchmark, dataset, metric, or evaluation protocol for robust RAG. | Must define evaluation axis, data, perturbation/threat, or metric. |

Table 62: Taxonomy codebook used for paper categorization. The full codebook is provided in the supplementary material.

| Pair | Rulebook verdict | External gold verdict | Gold-panel implication |
|---|---|---|---|
| P09 | Under-specified | Partial | Attack paper vs. defense paper within the same threat model should default to partial when failure type and evidence assumption align; the role, metric, and action differences make the papers complementary rather than directly rankable. |
| P12 | Partial | Partial | Verdict unchanged, but the panel refined the rationale: benchmark/detector vs. generation-method pairs should be partial when they share failure type and evidence assumption but diverge in intervention point, metric target, and action policy. |

Table 63: Rulebook revisions from independent gold adjudication. The panel labels are used as the external-gold reference for Table 57.

# F Core and Evidence Map Samples

| Artifact group | Representative files | Rows | Purpose |
|---|---|---|---|
| Evidence map and protocol | Adjudicated venue-expanded map, candidate expansion map, original adjudicated map, corpus-flow figure, canonical corpus table, search protocol, codebook, exclusion log | 342 adjudicated rows | Reproduce the canonical Map-B corpus, original Map-A adjudicated snapshot, coding fields, and inclusion/exclusion decisions. |
| Venue and source audit | `venue_systematic_queries.csv`, `venue_systematic_candidates.csv`, `venue_triage_decisions.csv` | 31 query rows / 51 candidates | Audit venue-linked recall risk and document triage decisions. |
| Venue-linked PDF collection | Venue-200 manifest, summary, builder script | 200 PDFs | Document the local venue-backed PDF collection used for follow-up coding; PDFs are local due to size. |
| Multimodal/scenario RAG collection | Multimodal-RAG-100 manifest, summary, builder script | 100 PDFs | Cover vision/multimodal, graph/KG, table/structured, agent/memory/tool, code/symbolic, domain-specific, and long-context RAG scenarios. PDFs are local due to size. |
| Human citation audit | Source-blind citation annotations, recommended/main-cite summaries, adjudication template | 300 papers | Records human decisions for which venue-linked and extension papers should be cited in the main text, case studies, appendix, background, or excluded. |
| Deep comparative cases | `deep_comparative_case_studies.csv`; main normalization findings table | 26 worksheet rows / 13 main cases | Compare assumptions, datasets, models, metrics, failure modes, and artifact availability for concrete normalization cases. |
| Expert-use validation | 12-pair pilot responses; 60-pair participant responses; 60-pair gold adjudications; disagreement/error summaries | 720 participant judgments / 192 gold labels | Reproduce the protocol-usability stress tests, expanded 60-pair validation, gold agreement, and coordinate-error analysis. |
| Map-B second-coding packet/results | Annotator packet, coordinator packet, second-coder completed labels, joint summary, disagreement CSV, adjudication worksheet, analysis script | 148 rows | Reports independent second coding over Map-B additions and queues disagreements for adjudication. |
| Expanded expert-validation packet | Protocol/control annotator packets, gold packet, coordinator packet, 60-pair pool, expertise template, analysis plan | 60 candidate pairs | Documents the design used for the completed 60-pair validation and supports future replications. |
| Action-labeled pilot | 60-item action-labeled pilot, three policy predictions, summary metrics, blueprint examples | 60 items / 180 predictions | Demonstrate answer/action/joint/unsafe metrics for action-control evaluation without claiming a deployed LLM leaderboard. |
| Expansion protocols | Venue-complete protocol; validation expansion protocol | protocol docs | Specify future manual venue-complete census and larger gold-adjudicated validation designs; these are agendas, not completed procedures in this paper. |
| Release manifest | `artifact_release_manifest.csv/json` | 303 artifacts | Provide versioned paths, row counts, and checksums for public release. |

Table 64: Release artifact manifest. The complete machine-readable manifest is provided as CSV/JSON in the supplement.

| Rank | Paper | Categories | Role |
|---|---|---|---|
| 1 | PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models | adv. corruption | attack |
| 2 | Certifiably Robust RAG against Retrieval Corruption | adv. corruption | defense |
| 3 | Making Retrieval-Augmented Language Models Robust to Irrelevant Context | noisy retrieval; distraction; calibration | method |
| 4 | Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training | noisy retrieval; evaluation | defense |
| 5 | Phantom: General Backdoor Attacks on Retrieval Augmented Language Generation | adv. corruption | attack |
| 6 | URAG: A Benchmark for Uncertainty Quantification in Retrieval-Augmented Large Language Models | OOD/uncertainty; calibration; evaluation; noise | benchmark |
| 7 | Trust or Abstain? A Self-Aware RAG Approach | knowledge conflict; calibration; evaluation | method |
| 8 | The First Drop of Ink: Nonlinear Impact of Misleading Information in Long-Context Reasoning | distraction; noisy retrieval | analysis |

Table 65: Sample entries from the core reading set. The full core set and metadata are provided in the supplementary material.

| Paper | Year | Failure / risk | Method family | Intervention | Boundary / note |
|---|---|---|---|---|---|
| PoisonedRAG | 2025 | adversarial corruption | attack | corpus / KB | Targeted knowledge corruption; stealth and long-term dynamics vary. |
| RobustRAG | 2024 | adversarial corruption | defense | retriever / aggregation | Certification assumes bounded corruption. |
| IrrelCtx | 2023 | retrieval noise / distraction | defense | context filtering | Does not model adaptive attackers. |
| RAAT | 2024 | retrieval noise | training | generator | Requires deployment-like noisy training data. |
| FaithfulRAG | 2025 | knowledge conflict | conflict modeling | generator | Depends on fact extraction and alignment. |
| AstuteRAG | 2025 | imperfect retrieval + conflict | control / consolidation | system control | Relies on source-aware LLM self-assessment. |
| Self-RAG | 2024 | retrieval need | adaptive retrieval | system control | Retrieval trigger calibration can shift. |
| RGB | 2024 | multi-axis RAG failure | benchmark | evaluation | Not adversarially adaptive. |
| RAGTruth | 2024 | hallucination / faithfulness | benchmark | evaluation | Strong word-level labels, less retrieval-threat focus. |
| MTRAG-UN | 2026 | unanswerability | benchmark | system action | Multi-turn setting; security is secondary. |

Table 66: Sample of the structured evidence map. The full CSV supplement contains title, authors, year, source, taxonomy categories, failure/risk type, method family, intervention point, benchmark/task, metrics, contribution, and limitation/note for the cited literature map.